

# PHYSICS READY

— SYSTEM 365 —

REAL PHYSICS.  
REAL CLASSROOMS.  
BRIGHTER  
FUTURES.

**VOLUME 4**  
OF 6

## WAVES, LIGHT & SOUND

Ready-to-Teach Secondary Physics

Observe • Model • Measure • Explain

**WAVES**  
TRANSFER ENERGY

**LIGHT**  
REVEALS MORE

**SOUND**  
CONNECTS  
OUR WORLD

WAVES  
LIGHT  
SOUND  
ELECTROMAGNETIC  
SPECTRUM

*SAME  
KNOWLEDGE.  
BRIGHTER  
TOMORROWS.*



FULL-YEAR  
TEACHING SYSTEM



40 READY-TO-USE  
LESSON PAGES



ENGAGING  
CLASSROOM ACTIVITIES



REAL-WORLD  
APPLICATIONS

**VOLUME 4 OF 6**

EMPOWERING TODAY'S STUDENTS  
FOR A BRIGHTER TOMORROW.

# CONTENTS

A COMPLETE LEARNING JOURNEY  
THROUGH WAVES, LIGHT & SOUND.

*Same knowledge.  
A brighter tomorrow.*



**WAVES**  
TRANSFER ENERGY.  
THE WORLD MOVES  
IN WAVES.

## 1 WAVE FUNDAMENTALS

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- 04 Transverse & Longitudinal Waves
- 05 Wave Anatomy
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- 07 Frequency & Period
- 08 The Wave Equation
- 09 Calculating Wave Speed
- 10 Wave Investigation



FOUNDATIONS  
FOR A WAVE WORLD.

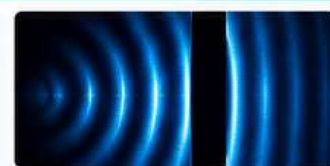


**WAVE BEHAVIOUR**  
SHAPES HOW ENERGY  
SPREADS.

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SAME WAVES.  
DIFFERENT PATHS.



**SOUND**  
CONNECTS  
OUR WORLD.

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FROM VIBRATIONS  
TO REAL APPLICATIONS.



**LIGHT**  
REVEALS  
MORE.

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LIGHT EXPLAINS  
THE WORLD WE SEE.

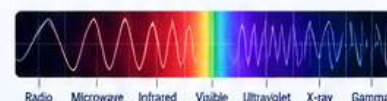


**ELECTROMAGNETIC  
WAVES CONNECT  
OUR LIVES.**

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FROM RADIO WAVES  
TO GAMMA RAYS.  
ONE SPECTRUM. A BRIGHTER FUTURE.



PHYSICS READY  
SYSTEM 365



COMPLETE  
TEACHING SYSTEM



READY-TO-USE  
LESSON PAGES



REAL-WORLD  
APPLICATIONS

**SAME KNOWLEDGE.  
A BRIGHTER TOMORROW.**

1 WAVE FUNDAMENTALS

CONCEPT LESSON 50-60 MIN

# ENERGY WITHOUT MATTER TRAVELLING WITH IT

Waves transfer  
energy, not  
matter.



## LEARNING OBJECTIVE:

Explain waves as mechanisms that transfer energy and information without overall transfer of matter.

## 1 2 CORE PHYSICS

- Waves transfer energy.
- Waves can transfer information.
- Particles may oscillate without travelling with the wave.
- Some waves require a medium.
- Electromagnetic waves can travel through a vacuum.

"A wave moves energy through a medium (or space), but the particles only oscillate around their positions."

## KEY VOCABULARY

- wave
- energy transfer
- medium
- oscillation
- electromagnetic wave
- vacuum

## 1 QUICK START

5 MIN

Compare the following examples of waves:



Water ripple



Sound



Light



Wave on a rope



## THINK ABOUT:

- What is the source?
- What type of wave is it?
- Where does the energy go?
- Does the material itself travel from the source to the destination?

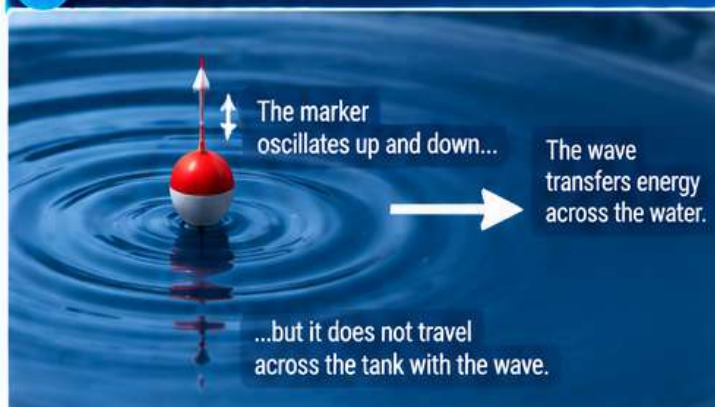
## 2 2 STUDENT TASK

For each example, identify:

Source → Wave → Energy Destination

| Example        | Source | Wave | Energy Destination |
|----------------|--------|------|--------------------|
| Water ripple   |        |      |                    |
| Sound          |        |      |                    |
| Light          |        |      |                    |
| Wave on a rope |        |      |                    |

## 3 3 VISUAL MODEL



## A COMMON MISCONCEPTION

"The material itself does not necessarily travel from source to receiver."

Particles in a medium (such as water, air or a rope) usually oscillate around their equilibrium position. They transfer energy to neighbouring particles, but they do not move with the wave.



## 5 CHALLENGE

Explain how a wave can transfer energy while the particles of a medium remain near their original positions.

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## 6 EXIT TICKET

What is transferred by a wave?

- (A) Matter
- (B) Energy
- (C) Particles
- (D) Mass



2 WAVE FUNDAMENTALS

COMPARE & EXPLAIN ⌚ 50–60 MIN

# TWO WAYS TO OSCILLATE

Waves can be transverse or longitudinal, depending on the direction of particle oscillation.

Same idea.  
Different direction.  
Different waves.



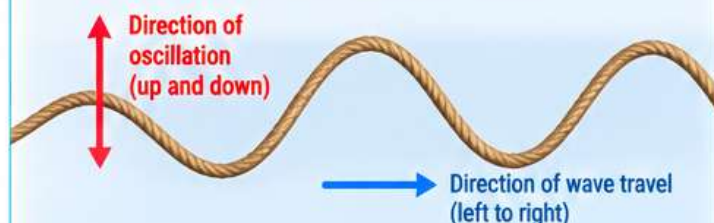
## LEARNING OBJECTIVE:

Distinguish transverse and longitudinal waves by comparing oscillation direction with energy-transfer direction.

## TRANSVERSE WAVES

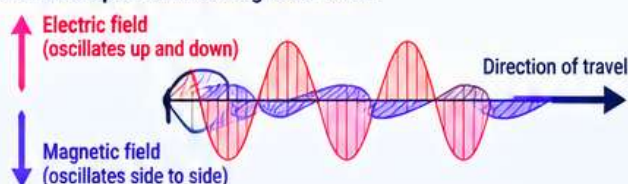
- Oscillations are perpendicular to the direction of wave travel.
- Examples:
  - electromagnetic waves
  - waves on a stretched string

OSCILLATION  
PERPENDICULAR  
TO TRAVEL



Example: wave on a stretched string

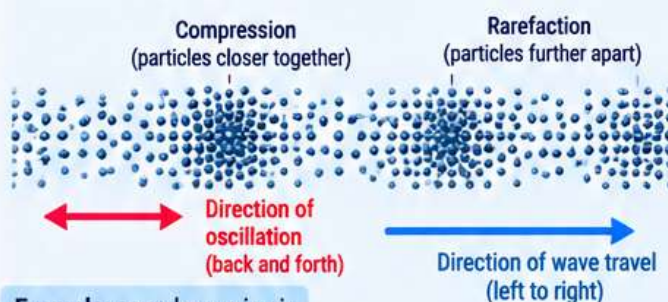
Other example: electromagnetic wave



## LONGITUDINAL WAVES

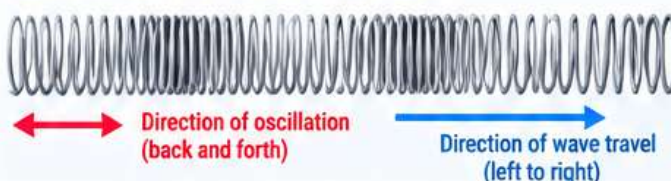
- Oscillations are parallel to the direction of wave travel.
- Example:
  - sound in air

OSCILLATION  
PARALLEL  
TO TRAVEL



Example: sound wave in air

Other example: longitudinal wave in a slinky



## 1 QUICK CHECK

Identify each wave as transverse (T) or longitudinal (L).

| Example                      | T or L? |
|------------------------------|---------|
| 1 Water wave on a pond       |         |
| 2 Sound in air               |         |
| 3 Wave on a stretched string |         |
| 4 Radio wave                 |         |
| 5 Wave in a slinky           |         |
| 6 Light wave                 |         |
| 7 Seismic P-wave (in Earth)  |         |
| 8 Seismic S-wave (in Earth)  |         |

## 2 STUDENT TASK

Complete the table below.

| Wave           | Direction of oscillation | Direction of travel |
|----------------|--------------------------|---------------------|
| Water ripple   |                          |                     |
| Sound wave     |                          |                     |
| Light wave     |                          |                     |
| Wave on a rope |                          |                     |

Use the diagrams above to help you.

## 3 CHALLENGE

Explain why drawing both transverse and longitudinal waves as simple wavy lines can create misconceptions.

## 4 THINK FURTHER

Can a wave be both transverse and longitudinal? Give an example and explain your answer.



## COMMON MISCONCEPTION

**"All waves look the same."**

Drawing longitudinal waves as a simple wavy line can make it look like the oscillation is up and down, which is incorrect.



## KEY TAKEAWAY

Wave type depends on the direction of particle oscillation compared to the direction of energy transfer.



3 WAVE FUNDAMENTALS

DIAGRAM LAB | ⌚ 50-60 MIN

# READ THE WAVE

Waves have a pattern.  
Each part tells a story.



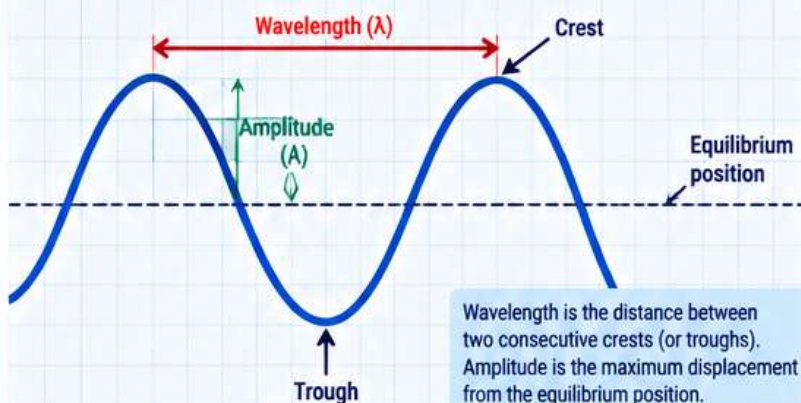
## LEARNING OBJECTIVE:

Identify and correctly use the principal features of a transverse wave.

### 1 KEY VOCABULARY

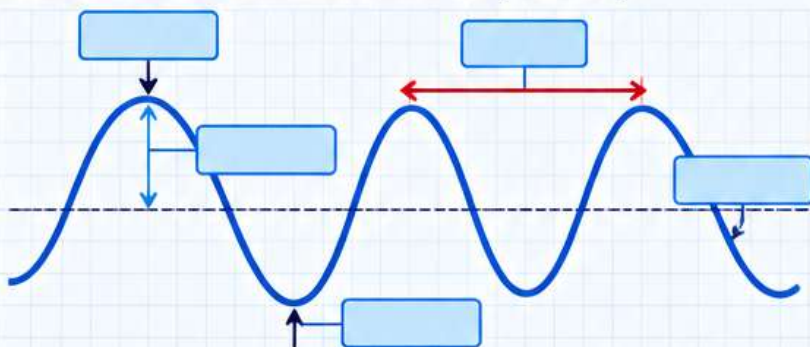
|                      |                                                               |
|----------------------|---------------------------------------------------------------|
| equilibrium position | The central resting position of the medium.                   |
| crest                | The highest point of the wave above the equilibrium position. |
| trough               | The lowest point of the wave below the equilibrium position.  |
| amplitude            | The maximum displacement from the equilibrium position.       |
| wavelength           | The distance between two successive crests (or troughs).      |

### 2 LABELLED WAVE DIAGRAM



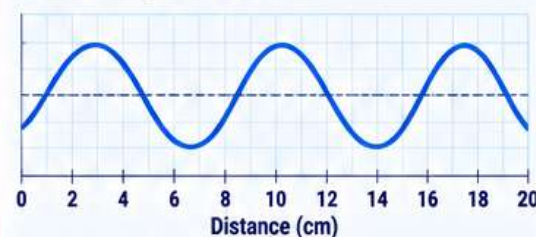
### 3 STUDENT TASK

Label the wave below. Use the words from the key vocabulary.



### 4 MEASURE IT

Use the scale below to determine the amplitude and wavelength of the wave.

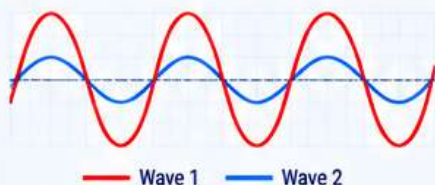


- a) Amplitude (A) = \_\_\_\_\_ cm  
b) Wavelength ( $\lambda$ ) = \_\_\_\_\_ cm

### 5 COMPARE

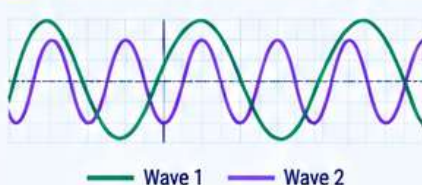
Look at the two waves and answer the questions.

#### A Same wavelength, different amplitude



- a) Which wave has the greater amplitude? \_\_\_\_\_  
b) What is the same about these waves? \_\_\_\_\_

#### B Same amplitude, different wavelength



- a) Which wave has the longer wavelength? \_\_\_\_\_  
b) What is the same about these waves? \_\_\_\_\_

### 6 CHALLENGE

Sketch a wave with twice the amplitude of the wave below, but with the same wavelength.



Now sketch your wave here:



#### COMMON MISCONCEPTION

A larger amplitude means a bigger transfer of energy, not a different speed or wavelength (in the same medium).



#### KEY TAKEAWAY

Waves can be fully described using a small number of key features.

3 WAVE FUNDAMENTALS

PRACTICAL SKILLS | 50–60 MIN

# FROM A DIAGRAM TO DATA



## LEARNING OBJECTIVE:

Measure wavelength and amplitude reliably from physical and graphical wave representations.

Careful measuring  
leads to accurate  
results.

## 1 CORE SKILLS

- identify equivalent points (e.g. crest to crest)
- measure multiple wavelengths
- divide to improve reliability
- measure from equilibrium position
- record units correctly



## KEY REMINDERS

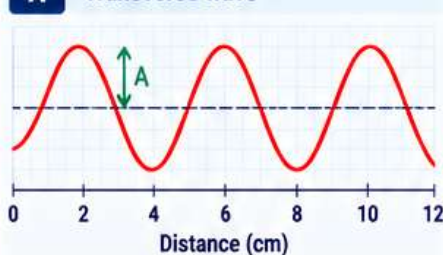
- ✓ Wavelength ( $\lambda$ ) is the distance between identical points (e.g. crest to crest).
- ✓ Amplitude ( $A$ ) is the maximum displacement from the equilibrium position.
- ✓ Use the scale on the diagram.
- ✓ Measure as accurately as possible.
- ✓ Always include units (m, cm, etc.).

**SAME POINTS.  
SAME DISTANCE.  
BETTER  
MEASUREMENTS.**

## 2 3 STUDENT TASK

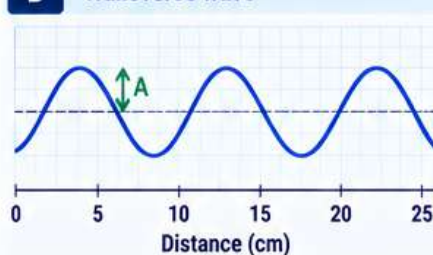
Measure the wave features in each diagram. Write your answers in the table.

### A Transverse wave



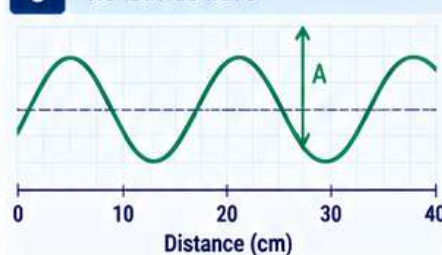
| Feature                  | Value |
|--------------------------|-------|
| Wavelength ( $\lambda$ ) |       |
| Amplitude ( $A$ )        |       |

### B Transverse wave



| Feature                  | Value |
|--------------------------|-------|
| Wavelength ( $\lambda$ ) |       |
| Amplitude ( $A$ )        |       |

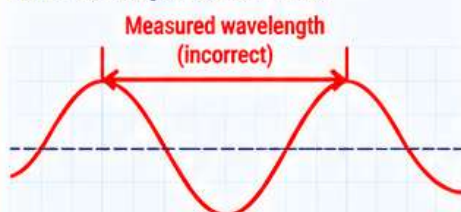
### C Transverse wave



| Feature                  | Value |
|--------------------------|-------|
| Wavelength ( $\lambda$ ) |       |
| Amplitude ( $A$ )        |       |

## 3 ERROR ANALYSIS

A student measures the wavelength from a crest to the next trough, as shown below.

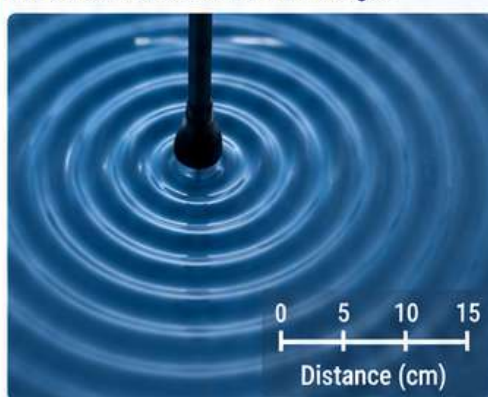


a) Why is this measurement incorrect?

b) Explain the correct method.

## 4 PRACTICAL APPLICATION

The image below shows water waves in a ripple tank. Use the scale to estimate the wavelength.



Estimated wavelength ( $\lambda$ ) = \_\_\_\_\_ cm

## 5 CHALLENGE

Explain why measuring five wavelengths and dividing by five may be more reliable than measuring just one.



## THINK ABOUT:

- measurement uncertainty
- reading the scale
- random errors
- the effect of small inaccuracies



3 WAVE FUNDAMENTALS

CONCEPT + CALCULATION | 40-50 MIN

# HOW OFTEN DOES A WAVE REPEAT?



## LEARNING OBJECTIVE:

Explain frequency and period and calculate one from the other.

Same wave.  
Different frequency.  
Different period.

### 1 KEY IDEAS

#### Frequency (f)

- Number of complete waves passing a point each second.
- Unit: **hertz (Hz)**
- Higher frequency = more waves every second.

#### Period (T)

- Time taken for one complete oscillation.
- Unit: **seconds (s)**
- Higher frequency = shorter period.

### 2 CORE RELATIONSHIP

$$T = 1 \div f$$

or

$$f = 1 \div T$$

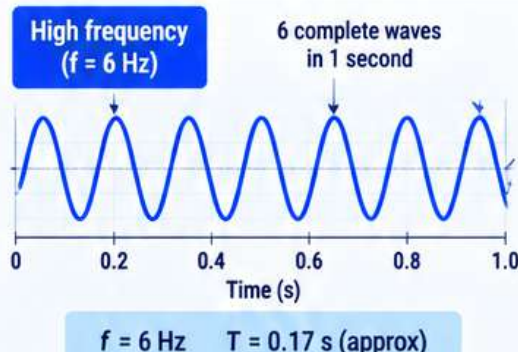
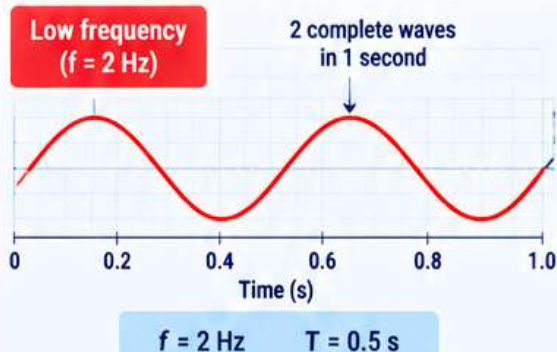
T = period (s)  
f = frequency (Hz)  
1 Hz = 1 s<sup>-1</sup>

### 3 UNITS AND MEANING

| Quantity  | Symbol | Unit        | Meaning                    |
|-----------|--------|-------------|----------------------------|
| Frequency | f      | hertz (Hz)  | Number of waves per second |
| Period    | T      | seconds (s) | Time for one complete wave |

A high frequency means the wave repeats quickly, so the period is small.  
A low frequency means the wave repeats more slowly, so the period is larger.

### 4 VISUAL COMPARISON Both waves are shown over the same time window (1 second).



**Key point:**  
Higher frequency means more waves in the same time, so a shorter period.

### 5 STUDENT TASK

Calculate the missing value in each case.

| Question | Frequency (Hz) | Period (s) |
|----------|----------------|------------|
| a)       | 5              | _____      |
| b)       | 20             | _____      |
| c)       | 0.5            | _____      |
| d)       | _____          | 0.25       |
| e)       | _____          | 0.1        |
| f)       | _____          | 2.0        |

### 6 REAL-WORLD EXAMPLES

| Example                      | Frequency (Hz) | Period (s) |
|------------------------------|----------------|------------|
| Human hearing (typical note) | 440            | 0.0023     |
| Car alarm sound              | 1000           | 0.001      |
| Ocean wave                   | 0.2            | 5.0        |
| Seismic wave (Earthquake)    | 1.0            | 1.0        |

Waves occur over a huge range of frequencies, from less than 1 Hz (earthquakes) to over 10<sup>14</sup> Hz (visible light).

### 7 CHALLENGE

Explain why increasing frequency decreases period.

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### 8 THINK FURTHER

A wave has a frequency of 8 Hz. How many complete waves pass a point in 30 seconds?

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3 WAVE FUNDAMENTALS

CALCULATION WORKSHOP ⌚ 40–50 MIN

# SPEED • FREQUENCY • WAVELENGTH

Three quantities.  
One equation.  
Many real-world applications.



## LEARNING OBJECTIVE:

Use the wave equation and explain the relationship between its quantities.

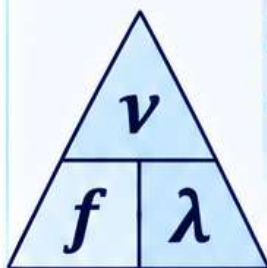
### 1 THE WAVE EQUATION

The speed, frequency and wavelength of a wave are related by:

$$v = f\lambda$$

$v$  = wave speed (m/s)  
 $f$  = frequency (Hz)  
 $\lambda$  = wavelength (m)

### 2 EQUATION TRIANGLE



How to use it:

- Cover the quantity you want to find.
- The remaining symbols show the operation.

$$v = f\lambda$$

$$f = v \div \lambda$$

$$\lambda = v \div f$$

### 3 UNITS CHECK

| Quantity   | Symbol    | Unit                           |
|------------|-----------|--------------------------------|
| Wave speed | $v$       | metres per second (m/s)        |
| Frequency  | $f$       | hertz (Hz or s <sup>-1</sup> ) |
| Wavelength | $\lambda$ | metres (m)                     |

✓ Always use consistent units (SI units).

### 4 WORKED EXAMPLE

A wave has a frequency of 5 Hz and a wavelength of 2 m. What is its speed?

Given:

$f = 5 \text{ Hz}$   
 $\lambda = 2 \text{ m}$

Use the wave equation:

$$v = f\lambda$$

$$v = 5 \times 2$$

$$v = 10 \text{ m/s}$$

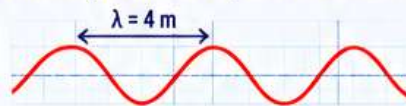
Answer:

The wave speed is **10 m/s**.

### 5 VISUAL EXAMPLE

Two waves travelling at the same speed in the same medium.

Wave A



$f = 2 \text{ Hz}$   
 $\lambda = 4 \text{ m}$   
 $v = 8 \text{ m/s}$

Wave B



$f = 4 \text{ Hz}$   
 $\lambda = 2 \text{ m}$   
 $v = 8 \text{ m/s}$

Both waves travel at the same speed (8 m/s), but Wave B has a higher frequency and a shorter wavelength.

### 6 STUDENT TASK

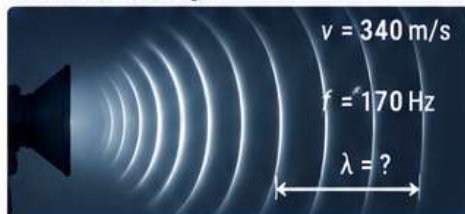
Calculate the missing value in each case. Show your working.

| #  | Given                                         | Find      | Working | Answer |
|----|-----------------------------------------------|-----------|---------|--------|
| a) | $f = 10 \text{ Hz}$ $\lambda = 3 \text{ m}$   | $v$       |         |        |
| b) | $v = 24 \text{ m/s}$ $\lambda = 6 \text{ m}$  | $f$       |         |        |
| c) | $v = 15 \text{ m/s}$ $f = 5 \text{ Hz}$       | $\lambda$ |         |        |
| d) | $f = 0.5 \text{ Hz}$ $\lambda = 20 \text{ m}$ | $v$       |         |        |
| e) | $v = 340 \text{ m/s}$ $f = 170 \text{ Hz}$    | $\lambda$ |         |        |



### REAL-WORLD CONTEXT

Sound travels in air at about 340 m/s. A sound has a frequency of 170 Hz. What is its wavelength?



### 7 CHALLENGE



Two waves travel at the same speed (12 m/s).

- Wave 1 has a frequency of 2 Hz. What is its wavelength?
- Wave 2 has a frequency of 6 Hz. What is its wavelength?
- Compare the two waves. How does the frequency affect the wavelength when the speed is the same?

### 8 THINK FURTHER

A water wave has a wavelength of 1.5 m and a frequency of 3 Hz. If the wave enters a different medium where its speed doubles, what happens to its wavelength? Explain.



### KEY TAKEAWAY

For a wave in the same medium, if the frequency increases, the wavelength decreases, and vice versa, as the speed remains constant.



3 WAVE FUNDAMENTALS

PROBLEM-SOLVING WORKSHOP | 40-60 MIN

# USE THE RIGHT INFORMATION

Same equation.  
Different contexts.  
Real-world physics.



## LEARNING OBJECTIVE:

Solve wave-speed problems in unfamiliar contexts.

### 1 THE CONTEXTS Waves travel in different media and are used in many situations.

#### WATER WAVES



- Mechanical wave
- Needs a medium (water)
- Typical speed: 0.1 – 10 m/s

#### SOUND



- Mechanical wave
- Needs a medium (air, water, solids)
- Typical speed in air: 340 m/s (at 20 °C)

#### RADIO WAVES



- Electromagnetic wave
- Can travel through a vacuum
- Speed in vacuum:  $3.0 \times 10^8$  m/s

#### ROPE WAVES



- Mechanical wave
- Needs a medium (rope)
- Typical speed: 1 – 50 m/s

### 2 PROBLEM-SOLVING STEPS

Follow these steps for each problem.

- 1 Identify known values.
- 2 Convert units where necessary.
- 3 Choose the correct equation ( $v = f\lambda$ ).
- 4 Calculate.
- 5 Attach the correct unit.
- 6 Interpret the result in context.

Same equation,  
new situations.  
Think. Convert.  
Calculate.  
Explain.

### 3 WORKED EXAMPLE

A water wave has a wavelength of 5.0 cm and a frequency of 4.0 Hz. What is its speed?

**Step 1:** Identify and convert units

$$\lambda = 5.0 \text{ cm} = 0.050 \text{ m}$$

$$f = 4.0 \text{ Hz}$$

**Step 2:** Choose equation

$$v = f\lambda$$

**Step 3:** Calculate

$$v = 4.0 \times 0.050$$

$$v = 0.20 \text{ m/s}$$

**Answer:**

The wave speed is **0.20 m/s**.



**Tip:**

Always convert to SI units (m, Hz, s) before calculating.

### 4 STUDENT TASK – MULTI-CONTEXT CALCULATIONS

Calculate the wave speed in each case. Show your working and final answer with units.

| #  | Context           | Wavelength ( $\lambda$ ) | Frequency (f) | Workings | Answer (with unit) |
|----|-------------------|--------------------------|---------------|----------|--------------------|
| a) | Water wave (lake) | 0.80 m                   | 2.5 Hz        |          |                    |
| b) | Sound (air)       | 0.50 m                   | 680 Hz        |          |                    |
| c) | Radio wave        | 3.0 m                    | 100 MHz       |          |                    |
| d) | Rope wave         | 0.25 m                   | 12 Hz         |          |                    |

### 5 MIXED CHALLENGE

A wave has a wavelength of 25 cm and a frequency of 3.0 kHz. Calculate its speed.

**Steps:**

- 1) Convert units.
- 2) Use  $v = f\lambda$ .
- 3) Calculate and give the answer with the correct unit.

### 6 PHYSICS REASONING

Two waves travel through the same medium at the same speed. Wave A has a frequency of 100 Hz and Wave B has a frequency of 400 Hz.

a) Which wave has the longer wavelength? Explain.

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b) If the speed is 340 m/s, calculate the wavelength of each wave.

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### 7 THINK FURTHER

In a different medium, the same wave has a higher speed but the same frequency. What happens to the wavelength? Explain.

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### KEY TAKEAWAY

- $v = f\lambda$
- Use the right information.
- Convert units.
- Attach units.
- Interpret your answer.
- The same equation works in many real-world contexts.



3 WAVE FUNDAMENTALS

PHYSICS LAB | ⌚ 40–60 MIN

# MEASURE A WAVE

Real measurements.  
Real physics.  
Real understanding.



## LEARNING OBJECTIVE:

Collect measurements to determine wave properties experimentally.

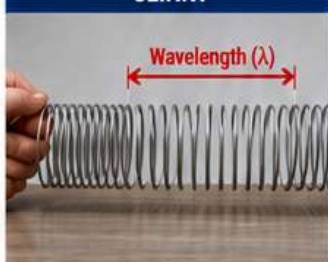
### 1 POSSIBLE SETUP Choose one of the following or a similar setup.

#### RIPPLE TANK



Water waves on the surface of a ripple tank.

#### SLINKY



Longitudinal waves in a slinky (compressions and rarefactions).

#### STRETCHED ROPE



Transverse waves on a stretched rope.

### 2 EQUIPMENT

You will need:



**Wave source**  
(e.g. motor, hand, or signal generator)



**Ruler**  
(to measure wavelength)



**Stopwatch**  
(to measure time)



**Reference markers**  
(e.g. tape, flags, or grid lines)



Use appropriate equipment and follow safety instructions.

### 3 INVESTIGATION STEPS

- 1 Set up your equipment.
- 2 Produce a steady, regular wave.
- 3 Measure the number of waves ( $N$ ) passing a point.
- 4 Measure the time ( $t$ ) for those waves.
- 5 Measure the wavelength ( $\lambda$ ).
- 6 Calculate the frequency:  $f = N \div t$
- 7 Calculate the wave speed:  $v = f\lambda$
- 8 Record your results in the table.

Measure.  
Calculate.  
Analyse.  
Repeat.

### 4 KEY FORMULAS

$$f = N \div t$$

$f$  = frequency (Hz)  
 $N$  = number of waves  
 $t$  = time (s)

$$v = f\lambda$$

$v$  = wave speed (m/s)  
 $f$  = frequency (Hz)  
 $\lambda$  = wavelength (m)

### 5 TIPS FOR ACCURATE RESULTS

- ✓ Use multiple wavelengths and divide.
- ✓ Keep the wave source steady.
- ✓ Measure from crest to crest (or identical points).
- ✓ Repeat each measurement at least 3 times.
- ✓ Use consistent units (SI units).
- ✓ Identify and reduce sources of uncertainty.

### 6 RESULTS TABLE

Complete the table with your measurements and calculated values.

| Trial   | Waves Counted ( $N$ ) | Time $t$ (s) | Frequency $f$ (Hz) | Wavelength $\lambda$ (m) | Wave Speed $v$ (m/s) |
|---------|-----------------------|--------------|--------------------|--------------------------|----------------------|
| 1       |                       |              |                    |                          |                      |
| 2       |                       |              |                    |                          |                      |
| 3       |                       |              |                    |                          |                      |
| Average |                       |              |                    |                          |                      |

### 7 ANALYSIS & EVALUATION

Answer the following questions.

- What are the main sources of uncertainty in your measurements?  
\_\_\_\_\_
- How could you improve the experiment?  
\_\_\_\_\_
- Why is it important to repeat measurements and calculate an average?  
\_\_\_\_\_

### 8 CHALLENGE



Compare your measured wave speed with a value calculated by another method. For example:

- Water wave: compare with  $v = \sqrt{gh}$  (shallow water approximation).
- Sound wave: compare with the known speed of sound in air ( $\approx 340$  m/s).
- Rope wave: compare with  $v = \sqrt{T/\mu}$  (tension  $T$ , linear density  $\mu$ ).

Discuss any differences. What might explain them?

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### KEY TAKEAWAYS

- ✓ Waves can be measured and analysed experimentally.
- ✓ Frequency, wavelength and wave speed are related by  $v = f\lambda$ .
- ✓ Accurate measurements require careful method and repetition.
- ✓ Experimental results can be compared with theoretical values.



3 WAVE BEHAVIOUR

VISUAL CONCEPT LESSON | ⌚ 20-30 MIN

# WAVES BOUNCE BACK

Same law.  
Different waves.  
Real-world applications.



## LEARNING OBJECTIVE:

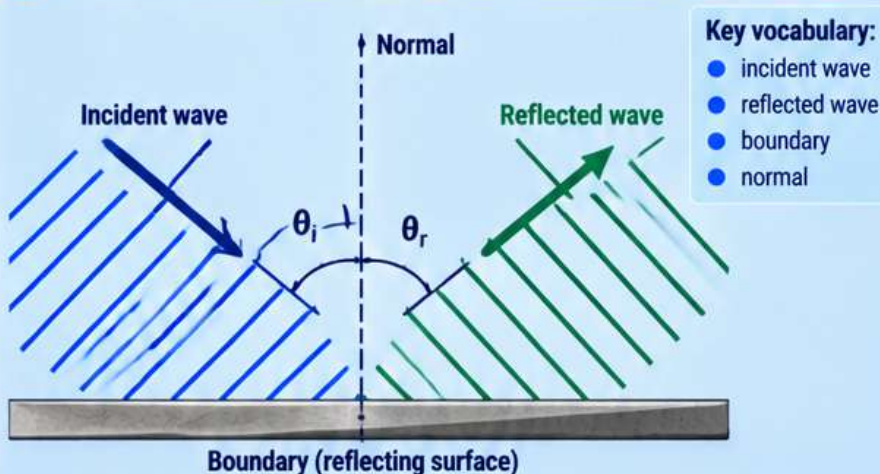
Describe wave reflection and identify incident and reflected waves.

### 1 KEY IDEAS

- When a wave reaches a boundary, it can bounce back. This is called **reflection**.
- The incoming wave is the **incident wave**.
- The outgoing wave is the **reflected wave**.
- The **normal** is an imaginary line perpendicular to the boundary.
- The **angle of incidence** equals the angle of reflection (for a flat boundary).

Angle of incidence = Angle of reflection  
 $\theta_i = \theta_r$

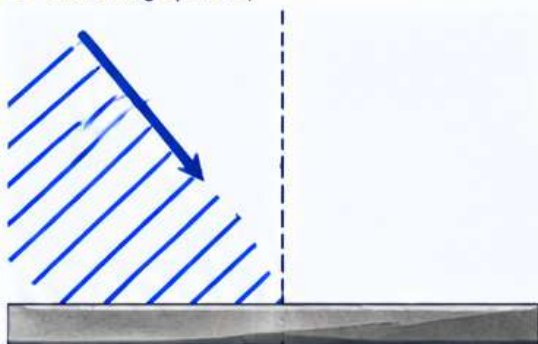
### 2 DIAGRAM: REFLECTION AT A STRAIGHT BARRIER



### 3 STUDENT TASK

Complete the diagram below by:

- drawing the **reflected wave**
- labeling the **incident wave**, reflected wave, normal and angles
- indicating  $\theta_i$  and  $\theta_r$



### 4 REAL-WORLD EXAMPLES



### 5 CHALLENGE

A plane wave hits a straight barrier at different angles.

- If the angle of incidence is  $30^\circ$ , what is the angle of reflection?  
\_\_\_\_\_
- If the angle of incidence is  $60^\circ$ , what is the angle of reflection?  
\_\_\_\_\_
- What happens to the direction of the reflected wave if the angle of incidence increases?  
\_\_\_\_\_  
\_\_\_\_\_



### 6 THINK FURTHER

Predict and explain:

If the incident angle increases, how does the direction of the reflected wave change? Why does this happen?

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### KEY TAKEAWAYS

- Waves can reflect at a boundary.
- Angle of incidence equals angle of reflection.
- The normal is perpendicular to the boundary.
- Reflection occurs for all types of waves (sound, light, water, radio, etc.).
- Reflection is important in many real-world technologies.



3 WAVE BEHAVIOUR

CONCEPT + DIAGRAM LAB | 20-30 MIN

# WHEN WAVES CHANGE SPEED



## LEARNING OBJECTIVE:

Explain refraction as a change in wave direction caused by a change in speed.

Different medium.  
Different speed.  
New direction.

### 1 KEY IDEAS

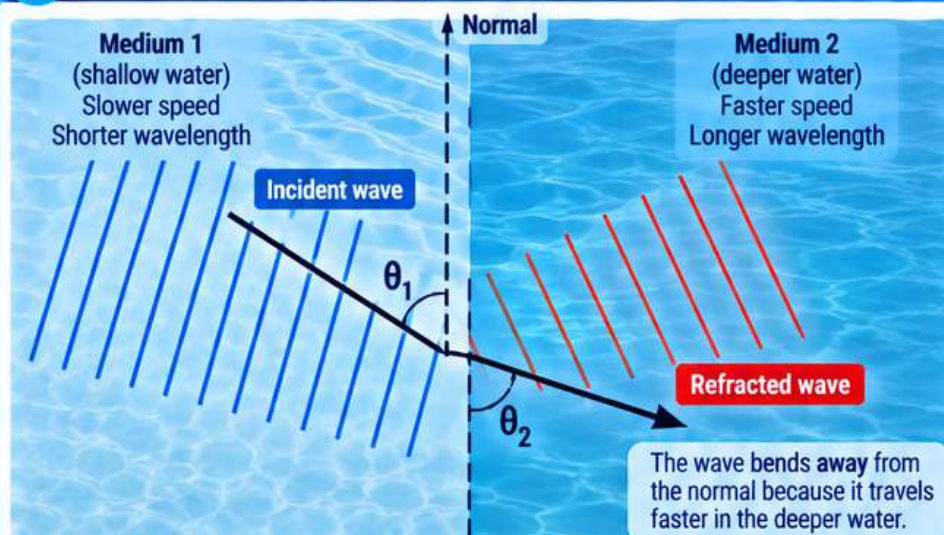
When a wave enters a different medium:

- its speed may change
- its wavelength may change
- its **frequency** remains determined by the source
- its **direction** may change (this is called **refraction**).



Refraction is the change in direction of a wave caused by a change in its speed.

### 2 DIAGRAM: REFRACTION AT A BOUNDARY



### 3 COMPARE BEFORE AND AFTER

Complete the table for the wave shown above. Write 'increases', 'decreases' or 'stays the same'.

| Quantity                 | Medium 1 (shallow) | Medium 2 (deep) |
|--------------------------|--------------------|-----------------|
| Speed ( $v$ )            | _____              | _____           |
| Wavelength ( $\lambda$ ) | _____              | _____           |
| Frequency ( $f$ )        | _____              | _____           |
| Direction                | _____              | _____           |

### 4 MORE EXAMPLES

**Water waves**

Shallow water (slower) | Deeper water (faster)

Waves bend away from the normal when they enter a faster medium.

**Sound waves**

Cool air (slower) | Warm air (faster)

Sound waves can refract in air when the temperature changes.

**Light waves**

Air (faster) | Glass (slower)

Light bends towards the normal when it enters a slower medium.

### 5 STUDENT TASK

A water wave has a frequency of 4 Hz and a wavelength of 3.0 cm in shallow water. It then enters deeper water where its speed increases.

- Does the frequency change? \_\_\_\_\_
- Does the wavelength increase or decrease? \_\_\_\_\_
- Does the wave bend towards or away from the normal? \_\_\_\_\_
- Explain your answers.

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### 6 CHALLENGE



A wave enters a new medium along the normal ( $\theta_1 = 0^\circ$ ). Its speed changes.

Explain why the wave does not change direction, even though its speed and wavelength may change.

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### 7 THINK FURTHER



In the diagram above, the wave bends away from the normal. What would happen if the wave entered a slower medium instead? How would the direction and wavelength change?

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### KEY TAKEAWAYS

- ✓ Refraction is a change in wave direction due to a change in speed.
- ✓ Speed and wavelength may change, but frequency remains the same.
- ✓ Waves bend towards the normal when they slow down and away from the normal when they speed up.
- ✓ If a wave enters along the normal, its direction does not change.



3 WAVE BEHAVIOUR

VISUAL INVESTIGATION | ⌚ 20–30 MIN

# WAVES SPREAD

Same waves.  
Different gaps.  
Different results.



## LEARNING OBJECTIVE:

Explain diffraction and relate the amount of spreading to wavelength and gap size.

### 1 WHAT IS DIFFRACTION?

- Diffraction is the spreading of waves when they **pass through** a gap or around an obstacle.
- The amount of spreading depends on the **wavelength** of the wave and the **size** of the gap or obstacle.



Waves spread out because different parts of the wavefront travel at different paths when passing through a gap or around a corner.

### 2 DIFFRACTION AT A GAP

Gap much larger than wavelength



Very little diffraction. Waves continue almost straight.

Gap size  $>$  wavelength  
( $d \gg \lambda$ )

Gap similar to wavelength



Waves spread out significantly after the gap.

Gap size  $\approx$  wavelength  
( $d \approx \lambda$ )

Gap much smaller than wavelength



Waves spread out a lot (after the gap they become almost circular).

Gap size  $<$  wavelength  
( $d \ll \lambda$ )

### 3 WAVELENGTH COMPARISON

Longer wavelength (low frequency)



Greater diffraction (more spreading).

Shorter wavelength (high frequency)



Less diffraction (less spreading).

### 4 REAL-WORLD EXAMPLES

Sound around doorways



Low-frequency sound can be heard around corners and through doorways.

Radio-wave reception



Radio-waves can diffract around buildings and hills, allowing reception even without a clear line of sight.

Water waves through gaps

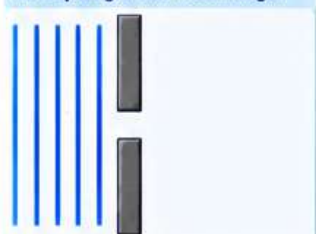


Water waves spread out after passing through a gap in a barrier.

### 5 STUDENT TASK

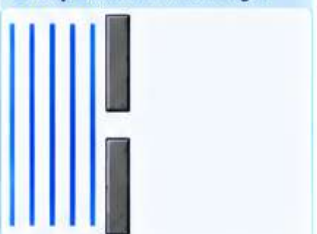
Predict the wavefront pattern after the gap. Draw your answer.

A. Gap larger than wavelength



Draw the wavefronts after the gap.

B. Gap similar to wavelength



Draw the wavefronts after the gap.

### 6 CHALLENGE



Low-frequency sound (longer wavelength) can often be heard around obstacles more effectively than very high-frequency sound (shorter wavelength).

Explain why.

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### 7 THINK FURTHER



A barrier blocks a direct path between a radio transmitter and a receiver, but the signal is still received.

What does this tell you about the wavelength of the radio waves? Explain.

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## KEY TAKEAWAYS

- ✓ Diffraction is the spreading of waves when they pass through a gap or around an obstacle.
- ✓ The amount of spreading depends on the wavelength and the size of the gap.
- ✓ Longer wavelengths diffract more than shorter wavelengths.
- ✓ Diffraction explains many real-world phenomena, including sound around corners and radio-wave reception.



3 WAVE BEHAVIOUR

PHYSICS MYSTERY | 30-40 MIN

# REFLECT, REFRACT OR DIFFRACT?



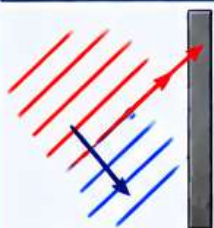
## LEARNING OBJECTIVE:

Identify wave behaviour from evidence and justify conclusions.

Look at the evidence.  
Think like a physicist.  
Explain your answer.

### 1 MYSTERY STATIONS For each station, analyse the diagram and information. Complete the table.

#### STATION 1 — SOLID BARRIER



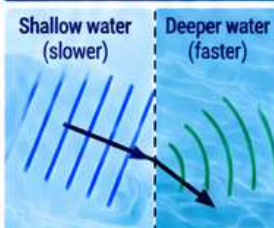
**Observation:**  
Waves hit a solid wall and bounce back. The angle of incidence equals the angle of reflection.

Wave behaviour

Evidence

Explanation

#### STATION 2 — NEW MEDIUM



**Observation:**  
Waves pass from shallow to deeper water and change direction. The wavelength increases.

Wave behaviour

Evidence

Explanation

#### STATION 3 — NARROW GAP



**Observation:**  
Waves pass through a narrow gap and spread out into a wider pattern beyond the gap.

Wave behaviour

Evidence

Explanation

#### 4 STATION 4 — OBSTACLE



**Observation:**  
Waves encounter an obstacle and spread out around the sides. Some waves are detected behind the obstacle.

Wave behaviour

Evidence

Explanation

#### 5 STATION 5 — ANGLED BOUNDARY



**Observation:**  
Waves hit a boundary at an angle and change direction towards the normal as they enter a slower medium.

Wave behaviour

Evidence

Explanation

#### 6 STATION 6 — MYSTERY CASE



**Observation:**  
Waves hit a barrier with a narrow gap. Some waves are reflected, and others pass through the gap and spread out.

Wave behaviour  
(1)

Wave behaviour  
(2)

Evidence

Explanation

#### 7 CHALLENGE

One of the stations shows two wave behaviours.

- Which station is it? \_\_\_\_\_
- Identify the two wave behaviours.
  - \_\_\_\_\_
  - \_\_\_\_\_
- Explain the sequence of events (which happens first? What happens next?).



#### 8 THINK FURTHER

A student says: "All waves bend when they hit something."

Do you agree? Use examples from the mystery stations to explain your answer.



#### KEY TAKEAWAYS

- ✓ Waves can reflect, refract or diffract depending on the situation.
- ✓ Reflection occurs at a boundary and the wave bounces back.
- ✓ Refraction occurs when the wave changes speed in a new medium.
- ✓ Diffraction occurs when the wave spreads out after passing through a gap or around an obstacle.
- ✓ More than one wave behaviour can occur in the same situation.



3 WAVE BEHAVIOUR

GRAPH SKILLS | 25–35 MIN

# WHAT DOES THE GRAPH SHOW?



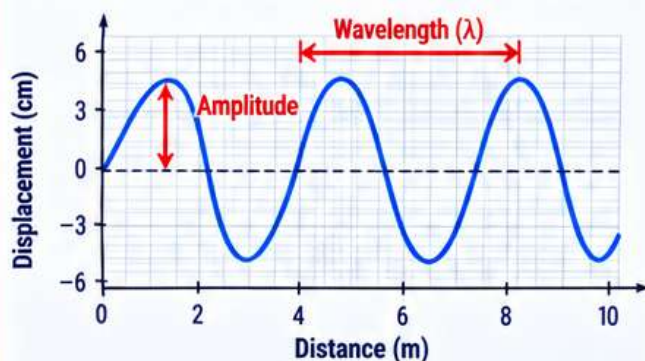
## LEARNING OBJECTIVE:

Extract amplitude, wavelength, period and frequency information from wave graphs.

Read the graph.  
Find the information.  
Use the physics.  
Explain your thinking.

### 1 DISPLACEMENT–DISTANCE GRAPH

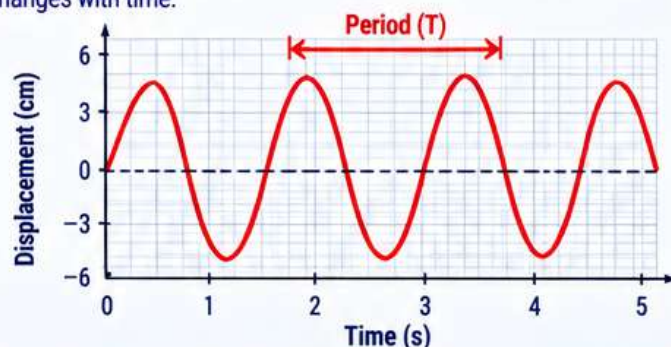
This graph shows a snapshot of a wave at a particular time.



- What is the amplitude of the wave? \_\_\_\_\_
- What is the wavelength of the wave? \_\_\_\_\_
- How many complete wavelengths are shown in the graph? \_\_\_\_\_

### 2 DISPLACEMENT–TIME GRAPH

This graph shows how the displacement of a point on the wave changes with time.



- What is the amplitude of the wave? \_\_\_\_\_
- What is the period of the wave? \_\_\_\_\_
- Calculate the frequency using  $f = \frac{1}{T}$ . \_\_\_\_\_
- How many complete cycles are shown in the graph? \_\_\_\_\_

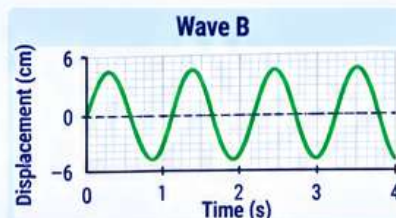
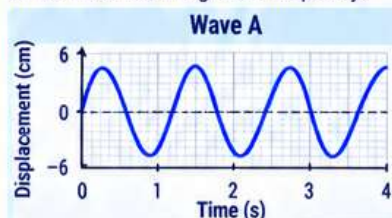
### 3 CALCULATIONS FROM GRAPHS

Use the information from the graphs to complete the table.

| Quantity   | Symbol    | Value | Unit |
|------------|-----------|-------|------|
| Amplitude  | $A$       |       | cm   |
| Wavelength | $\lambda$ |       | m    |
| Period     | $T$       |       | s    |
| Frequency  | $f$       |       | Hz   |

### 4 TWO WAVES FOR COMPARISON

Which wave has the greater frequency?



- Which wave has the greater frequency? \_\_\_\_\_
- Explain how you know. \_\_\_\_\_

### 5 COMMON MISCONCEPTION



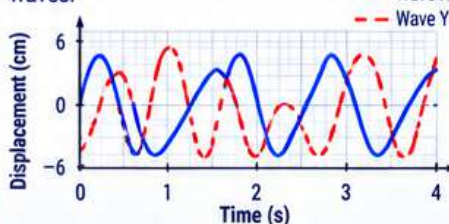
"You can read the wavelength directly from a displacement–time graph."

**FALSE**

A displacement–time graph shows how a point on the wave moves with time. It does not show the distance between points on the wave, so the wavelength cannot normally be read directly from this graph without additional information (such as the wave speed).

### 6 CHALLENGE

The graphs below show two different waves.



- Which wave has the greater frequency? \_\_\_\_\_
- Which wave has the greater period? \_\_\_\_\_
- Explain your reasoning. \_\_\_\_\_



### KEY TAKEAWAYS

- Amplitude is the maximum displacement from the equilibrium position.
- Wavelength can be read from a displacement–distance graph.
- Period can be read from a displacement–time graph.
- Frequency is calculated using  $f = \frac{1}{T}$ .
- You cannot normally read the wavelength directly from a displacement–time graph.
- Compare graphs carefully to determine which wave has the greater frequency.



3 WAVE BEHAVIOUR

DATA DETECTIVE | 50-60 MIN

# FIND THE PATTERN



## LEARNING OBJECTIVE:

Analyse wave data and test relationships between frequency, wavelength and speed.

Same medium.  
Different waves.  
What do you notice?

### 1 CONTEXT

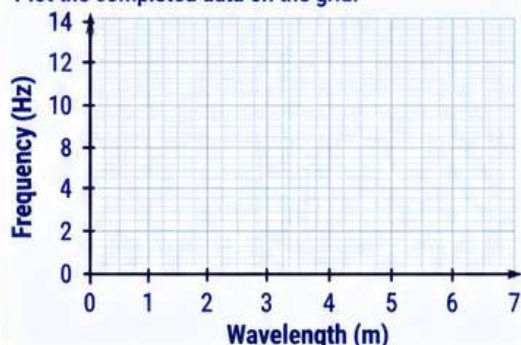
A group of waves travel through the **same** medium. The table shows measured frequency and wavelength values. Some data is missing. Your task is to **complete the table**, look for **patterns** and analyse the results.

### 3 STUDENT MISSION

- 1 Calculate the missing values using the wave equation  $v = f\lambda$ .
- 2 Identify the **pattern** between frequency and wavelength.
- 3 Find the anomalous result (if any). Explain your reasoning.
- 4 Plot a graph of frequency against wavelength using the completed data.
- 5 Explain why the relationship is not linear.
- 6 Evaluate whether the wave speed is approximately constant for this medium.

### 4 GRAPH: FREQUENCY vs WAVELENGTH

Plot the completed data on the grid.



### 2 DATASET: WAVES IN THE SAME MEDIUM

| Wave | Frequency $f$ (Hz) | Wavelength $\lambda$ (m) | Wave Speed $v$ (m/s) |
|------|--------------------|--------------------------|----------------------|
| A    | 2                  | 6.0                      |                      |
| B    | 4                  | 3.0                      |                      |
| C    | 6                  | 2.0                      |                      |
| D    | 10                 |                          | 20                   |
| E    | 5                  |                          | 20                   |
| F    |                    | 1.0                      | 20                   |
| G    | 8                  | 2.5                      |                      |
| H    | 12                 | 1.8                      |                      |
| I    | 3                  |                          | 19                   |
| J    | 7                  | 2.9                      |                      |

### 5 ANALYSIS QUESTIONS

- What pattern do you notice between frequency and wavelength for waves in the same medium?  
\_\_\_\_\_
- Why is the relationship not linear? Explain using the wave equation.  
\_\_\_\_\_
- Which result is anomalous? \_\_\_\_\_ Why do you think this?  
\_\_\_\_\_
- Is the wave speed approximately constant for this medium? Give evidence from your data.  
\_\_\_\_\_

### 6 CHALLENGE

The wave speed in this medium is approximately 20 m/s. Predict the wavelength for a wave with a frequency of 15 Hz. Show your working and explain your answer.

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### 7 EXTENSION

A wave in the same medium has a wavelength of 0.5 m. What is its frequency?  
How does this compare to the trend in your graph?  
Would you expect this wave to have a higher or lower frequency than the waves in the table? Explain.

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### KEY TAKEAWAYS

- ✓ For a given medium, wave speed is approximately constant.
- ✓ As frequency increases, wavelength decreases.
- ✓ The relationship between frequency and wavelength is inversely proportional ( $v = f\lambda$ ).
- ✓ Data can be used to identify patterns and anomalous results.
- ✓ Graphs help to visualise relationships in wave data.



3 SOUND

CONCEPT LESSON | 40-50 MIN

# VIBRATIONS YOU CAN HEAR

Sound is a wave  
that needs a medium  
to travel.



## LEARNING OBJECTIVE:

Explain how sound is produced and transmitted as a longitudinal wave.

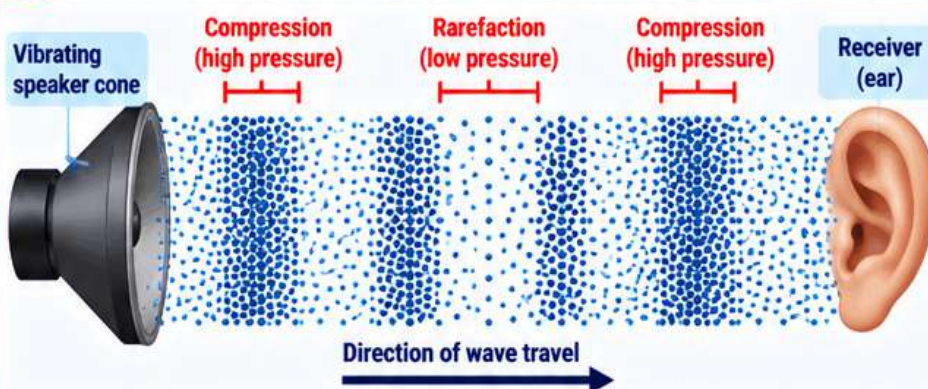
### 1 HOW SOUND IS PRODUCED

Sound is produced by a **vibrating source**. When an object vibrates, it pushes and pulls the particles in the surrounding medium (such as air), creating a longitudinal wave.



A **longitudinal wave** has **vibrations parallel** to the direction of wave travel.

### 2 SOUND WAVE IN AIR



### 3 PARTICLE OSCILLATION

Particles in the medium do not travel with the wave. They oscillate back and forth around their equilibrium position.

**Particle oscillation**  
(parallel to wave direction)



Compression  
(particles close together)

Rarefaction  
(particles further apart)

### 4 KEY FEATURES OF SOUND



**Compression**  
Region where particles are close together. Higher pressure.



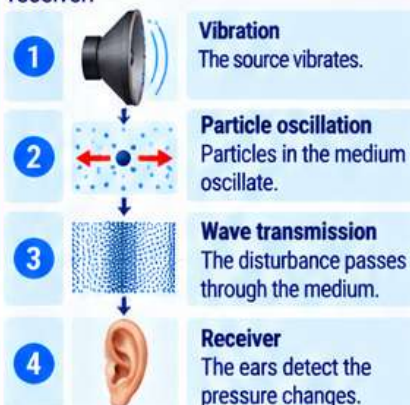
**Rarefaction**  
Region where particles are further apart. Lower pressure.



**Longitudinal oscillation**  
Particles vibrate parallel to the direction of wave travel.

### 5 FROM SOURCE TO RECEIVER

Complete the sequence to show how sound travels from the source to the receiver.



### 6 STUDENT TASK

- Label the compressions and rarefactions in the diagram in section 2.
- Explain, in your own words, how the vibrating speaker produces a sound wave in air.
- Why do particles in air not travel with the wave?
- How is the direction of particle oscillation different from the direction of wave travel?

### 7 CHALLENGE

Explain why sound cannot travel through a vacuum.

#### Think further:

How would a sound wave be different in a solid compared to a gas?



### KEY TAKEAWAYS

- ✓ Sound is produced by a vibrating source.
- ✓ Sound travels as a longitudinal wave.
- ✓ Compressions are high-pressure regions; rarefactions are low-pressure regions.
- ✓ Particles oscillate parallel to the direction of wave travel.
- ✓ Sound needs a medium to travel and cannot travel through a vacuum.



3 SOUND

COMPARE & EXPLAIN | 40-50 MIN

# WHY DO SOUNDS HAVE DIFFERENT PITCHES?



## LEARNING OBJECTIVE:

Relate sound frequency to perceived pitch.

Same medium.  
Different frequency.  
Different pitch.

### 1 WHAT IS PITCH?

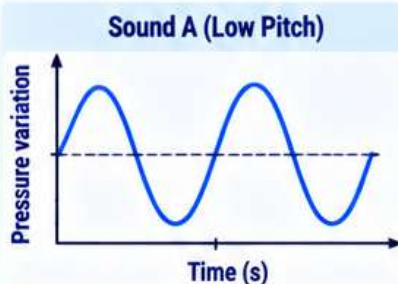
Pitch is how high or low a sound seems. It depends on the frequency of the sound wave (the number of complete vibrations per second).



Higher frequency → higher pitch  
Lower frequency → lower pitch

### 2 COMPARING SOUND WAVES

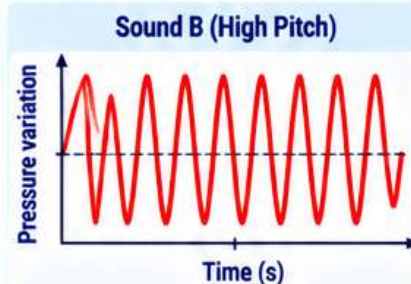
Two sound waves are shown over the same time interval (0.020 s).



Time interval: 0.020 s

Number of cycles: 2

Frequency: \_\_\_\_\_ Hz



Time interval: 0.020 s

Number of cycles: 8

Frequency: \_\_\_\_\_ Hz

### 3 STUDENT TASK

a) Calculate the frequency of sound A and sound B

$$\text{using } f = \frac{\text{number of cycles}}{\text{time}}$$

b) Which sound has the higher pitch? \_\_\_\_\_

c) Rank the following sounds from lowest pitch (1) to highest pitch (4).

| Sound                         | Rank (1-4) |
|-------------------------------|------------|
| Alarm siren (high)            | _____      |
| Bass guitar note (low)        | _____      |
| Human voice (adult male, low) | _____      |
| Flute note (high)             | _____      |

### 4 REAL-WORLD EXAMPLES

#### Musical instruments



Thicker strings vibrate at lower frequencies (lower pitch). Thinner strings vibrate at higher frequencies (higher pitch).

#### Human voice



A deeper voice has a lower frequency. A higher-pitched voice has a higher frequency.

#### Alarms and signals



Emergency alarms use high-frequency sounds, which we hear as high-pitched.

#### Animals



Dogs can hear higher-pitched sounds than humans. Bats use very high-frequency sounds (ultrasound).

### 5 CHALLENGE



Explain why changing the frequency of a sound wave changes its pitch without necessarily changing its loudness.

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### 6 THINK FURTHER



Two sounds have the same pitch but different loudness. What is the same about their waves? What is different?

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### KEY TAKEAWAYS

- ✓ Pitch is determined by frequency.
- ✓ Higher frequency means higher pitch.
- ✓ Lower frequency means lower pitch.
- ✓ Pitch and loudness are different properties.
- ✓ Many real-world sounds have characteristic pitch ranges.



3 SOUND

VISUAL CONCEPT | 40-50 MIN

# HOW STRONG IS THE VIBRATION?



## LEARNING OBJECTIVE:

Relate wave amplitude to sound intensity and perceived loudness.

Same pitch.  
Different loudness.  
What changes?

### 1 AMPLITUDE AND LOUDNESS

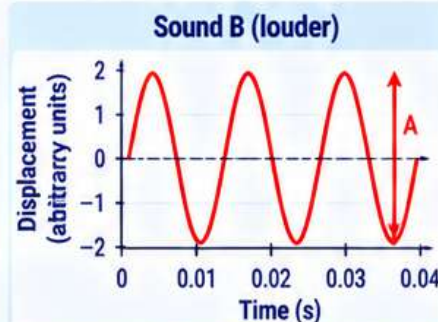
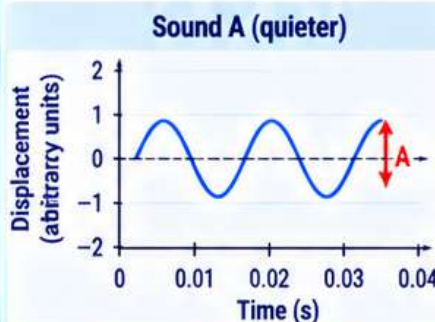
Loudness depends on the amplitude of the sound wave. A larger amplitude means a more intense vibration, which we hear as a louder sound.



**Amplitude** is the **maximum displacement** from the equilibrium position.

### 2 SAME FREQUENCY, DIFFERENT AMPLITUDES

Both waves have the same frequency (same pitch) but different amplitudes.



### 3 STUDENT TASK

Look at the waveforms above and answer:

- Which sound is louder? \_\_\_\_\_
- Do the sounds have the same pitch? Explain. \_\_\_\_\_
- What is different between the two waves? \_\_\_\_\_
- Estimate the amplitude of each wave (from the graph).

Sound A: \_\_\_\_\_ Sound B: \_\_\_\_\_

### 4 AMPLITUDE vs FREQUENCY

| Property                  | Amplitude (A)              | Frequency (f)                        |
|---------------------------|----------------------------|--------------------------------------|
| What it changes           | Loudness (sound intensity) | Pitch (how high or low the sound is) |
| What it does not change   | Pitch                      | Loudness (usually)                   |
| Unit                      | metres (m)                 | hertz (Hz)                           |
| Visual effect on waveform | Taller or shorter wave     | More or fewer waves in the same time |

### 5 COMMON MISCONCEPTION



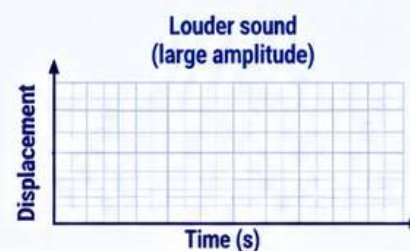
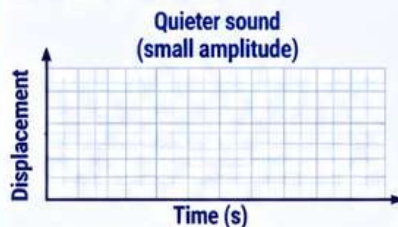
"A taller waveform means a higher pitch."

**FALSE**

A taller waveform means a larger amplitude, which is a louder sound. Pitch depends on frequency (the number of cycles per second), not amplitude.

### 6 CHALLENGE

Sketch two sound waves with the same pitch but different loudness.



### 7 REAL-WORLD EXAMPLES

#### Music



Hitting the drum harder produces a larger amplitude and a louder sound, but the pitch stays the same.

#### Everyday sounds



Turning up the volume on a speaker increases the amplitude, making the sound louder.

#### Nature



Thunder can be very loud because it has a large amplitude, but its pitch depends on the frequency of the sound waves.



#### KEY TAKEAWAYS

- Loudness is related to the amplitude of the sound wave.
- A larger amplitude means a louder sound.
- Pitch is related to frequency, not amplitude.
- Sounds can have the same pitch but different loudness.
- A taller waveform does not mean a higher pitch.



3 SOUND

PRACTICAL INVESTIGATION | 50-60 MIN

# SEE THE SOUND



## LEARNING OBJECTIVE:

Use sound data to determine frequency and compare waveforms.

Different sounds.  
Different waveforms.  
What can you find?

### 1 EQUIPMENT / DATA SOURCE

Use a microphone connected to sound visualisation software, or the supplied oscilloscope traces.



Microphone

Converts sound into an electrical signal.



Sound visualization software

Shows the waveform on a screen.



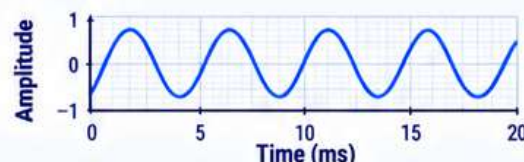
Sound source

Examples: musical instrument, tuning app, clapping, voice or other noise source.

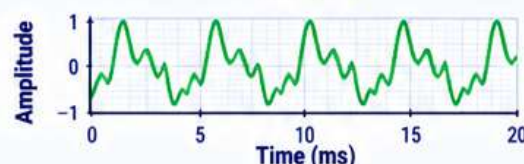
### 2 EXAMPLE WAVEFORMS

These are waveforms of three different sounds recorded using a microphone and software.

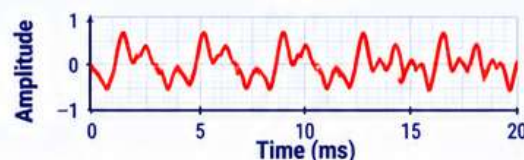
Sound A  
(tuning fork)



Sound B  
(guitar note)



Sound C  
(human voice)



### 3 STUDENT TASK

- 1 Observe the waveform for each sound.
- 2 Measure the period (time for one complete cycle).
- 3 Calculate the frequency using  $f = \frac{1}{T}$ .
- 4 Compare the waveforms and describe the differences in shape and amplitude.

### 4 RESULTS TABLE

Complete the table using your measurements.

| Source                                    | Period, $T$<br>(ms) | Frequency,<br>$f$ (Hz) | Relative Amplitude<br>(low / medium / high) | Observation<br>(e.g. waveform shape) |
|-------------------------------------------|---------------------|------------------------|---------------------------------------------|--------------------------------------|
| Sound A<br>(tuning fork)                  | _____               | _____                  | _____                                       | _____                                |
| Sound B<br>(guitar note)                  | _____               | _____                  | _____                                       | _____                                |
| Sound C<br>(human voice)                  | _____               | _____                  | _____                                       | _____                                |
| Your own sound<br>(e.g. clap, instrument) | _____               | _____                  | _____                                       | _____                                |

### 5 ANALYSIS QUESTIONS

- a) Which sound has the highest frequency? \_\_\_\_\_
- b) Which sound has the greatest amplitude? \_\_\_\_\_
- c) How do the waveform shapes differ between the sounds?  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
- d) Which sound has the most complex waveform? \_\_\_\_\_
- e) What might cause the differences in waveform shape?  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

### 6 CHALLENGE

Identify two different sounds that produce similar frequencies but different waveform shapes.

Name the sounds: \_\_\_\_\_

How do you know their frequencies are similar? \_\_\_\_\_

How do their waveforms differ?  
\_\_\_\_\_  
\_\_\_\_\_



### KEY TAKEAWAYS

- A microphone converts sound into an electrical signal.
- The period can be measured from a waveform.
- Frequency is calculated using  $f = \frac{1}{T}$ .
- Different sounds can have the same frequency but different waveform shapes.
- Waveform shape depends on the sound source and the combination of frequencies present.
- Sound visualisation helps us see and analyse sound waves.



3 SOUND

COMPARE & EXPLAIN | 50-60 MIN

# DOES SOUND TRAVEL THE SAME EVERYWHERE?



## LEARNING OBJECTIVE:

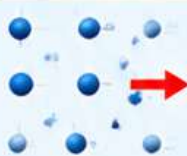
Explain why sound speed differs between materials.

Same sound.  
Different materials.  
Different speeds.

### 1 HOW SOUND TRAVELS IN DIFFERENT MATERIALS

Sound travels through a medium by particles vibrating and passing the disturbance to neighbouring particles. The closer the particles and the stronger their interactions, the faster the sound travels.

#### GAS (air)



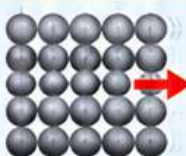
Particles are far apart.  
Weaker interactions.  
Slower sound speed.

#### LIQUID (water)



Particles are closer together. Stronger interactions.  
Faster than in gases.

#### SOLID (metal)



Particles are tightly packed. Strongest interactions.  
Fastest sound speed.

### 2 SOUND SPEED DATA

Approximate speeds of sound in different materials (at around 20 °C).

| Material          | Speed of sound (m/s) |
|-------------------|----------------------|
| Air (gas)         | 343                  |
| Helium (gas)      | 972                  |
| Water (liquid)    | 1,480                |
| Seawater (liquid) | 1,530                |
| Wood (solid)      | 3,300                |
| Glass (solid)     | 5,640                |
| Steel (solid)     | 5,960                |
| Aluminium (solid) | 6,320                |

Faster in solids

Slower in gases

### 3 STUDENT TASK

- Use the data table to rank the materials from slowest to fastest sound speed.
- Identify any patterns. What do you notice?
- Compare the speeds in gases, liquids and solids.
- Explain why sound generally travels more rapidly through solids than gases.
- Give a real-world example for each state of matter where sound travel is important.

### 4 REAL-WORLD EXAMPLES

#### Gases



We hear thunder (in air) a few seconds after we see lightning because sound travels relatively slowly in air.

#### Liquids



Sound travels much faster in water, allowing whales to communicate over long distances.

#### Solids



Vibrations travel very quickly through solid materials, such as rails, which is why you can sometimes hear a train before you see it.

### 5 CHALLENGE: CALCULATE TRAVEL TIME

A sound pulse travels a distance of 1,000 m through two different materials. Use the data table to calculate the time taken in each case.

- a) Through air (343 m/s)

$t = \underline{\hspace{2cm}} \text{ s}$

- b) Through steel (5,960 m/s)

$t = \underline{\hspace{2cm}} \text{ s}$

- c) How many times faster does the sound travel in steel compared to air?

$\underline{\hspace{2cm}}$  times faster

### 6 THINK FURTHER

- a) Why do you think sound travels faster in liquids than in gases?

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- b) Would you expect sound to travel faster in ice or in liquid water? Explain your reasoning.

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#### KEY TAKEAWAYS

- Sound speed depends on the properties of the medium.
- Particles are closer together in solids, so vibrations pass more quickly.
- Sound generally travels fastest in solids, slower in liquids and slowest in gases.
- Different materials have different sound speeds.
- Sound speed affects real-world situations such as communication, detection and timing.



3 SOUND

REAL-WORLD PHYSICS 50-60 MIN

# WHEN SOUND RETURNS

Sound doesn't always stop. Sometimes it comes back!



## LEARNING OBJECTIVE:

Explain echoes as reflected sound waves.

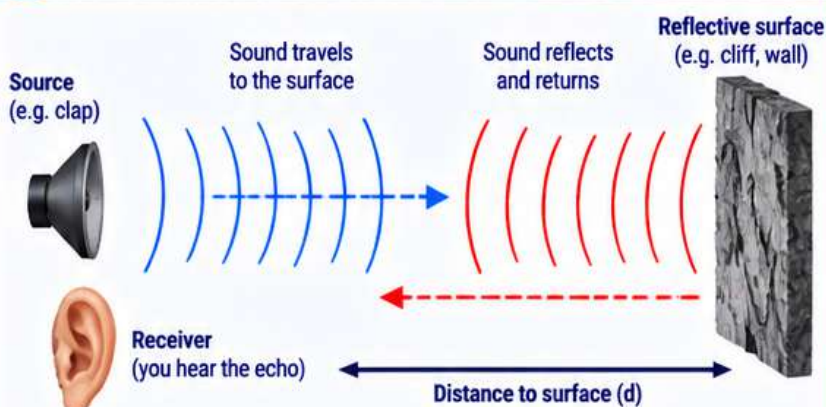
### 1 HOW ECHOES HAPPEN

An echo is a reflected sound wave. Sound travels from a source, hits a surface, reflects and returns to the receiver. If the reflected sound arrives at least 0.1 seconds after the original sound, we hear it as a separate echo.



**Echoes are caused by reflection of sound waves from a surface.**

### 2 THE ECHO PROCESS



### 3 WHEN WILL YOU HEAR AN ECHO?

You are more likely to hear a noticeable echo when:

- ✓ The surface is large, hard and smooth (e.g. rock face, building wall).
- ✓ You are a long distance from the surface (typically more than 17 m).
- ✓ The environment is quiet.
- ✓ The sound is short and sharp (e.g. a clap).
- ✓ The surface reflects sound well (e.g. concrete, rock, metal).

### 4 REAL-WORLD APPLICATIONS

#### Sonar



Ships and submarines use echoes (SONAR) to detect objects and measure distance underwater.

#### Distance measurement



Bats use echoes to find their surroundings. The time taken for the echo to return can be used to calculate distance.

#### Acoustic design



Recording studios and theatres use materials that absorb sound to reduce unwanted reflections and create clearer sound.

### 5 STUDENT TASK

a) Which of the following situations are likely to produce a noticeable echo? Tick (✓) the most likely ones.

- ☐ A clap in a small classroom
- ☐ A shout in a large empty gym
- ☐ A whistle near a mountain cliff
- ☐ A car horn in a city street
- ☐ A clap in a carpeted room

b) Explain your choices for two of the situations.

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### 6 CHALLENGE



Soft materials (e.g. curtains, carpets, acoustic foam) are commonly used in recording spaces. Explain why these materials help to reduce unwanted reflections.

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### KEY TAKEAWAYS

- ✓ Echoes are reflected sound waves.
- ✓ You hear a separate echo if the delay is at least 0.1 seconds.
- ✓ Large, hard and smooth surfaces reflect sound well.
- ✓ Echoes are used in sonar, distance measurement and acoustic design.
- ✓ Soft materials absorb sound and reduce reflections.



3 SOUND

CALCULATION WORKSHOP | 50-60 MIN

# THERE AND BACK AGAIN

The sound travels to the surface and back. Don't forget to divide by two!



## LEARNING OBJECTIVE:

Calculate distance using sound speed and echo time.

### 1 CORE RELATIONSHIP

Distance is calculated using the wave equation:

$$\text{distance} = \text{speed} \times \text{time}$$

For an echo, the sound travels to the surface and back, so the measured time is for the total path.

$$\text{Distance to surface} = \frac{\text{speed} \times \text{time}}{2}$$

### 2 WORKED EXAMPLE

A sonar pulse is sent towards a cliff. The echo is detected 0.40 s later. The speed of sound in air is 340 m/s.

**Step 1:** Calculate total path length

$$\text{total path} = 340 \text{ m/s} \times 0.40 \text{ s} = 136 \text{ m}$$

**Step 2:** Find distance to surface

$$\text{distance} = 136 \text{ m} \div 2 = 68 \text{ m}$$

**Answer:** The cliff is 68 m away.

### 3 VISUAL EXPLANATION

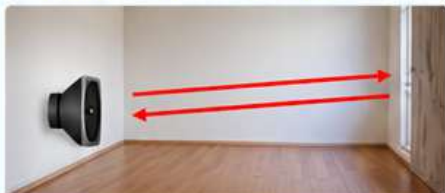


### 4 STUDENT TASK

Calculate the distance to the surface in each situation. Show your working.

#### a) Room (air)

Echo time = 0.25 s  
Speed of sound = 340 m/s



Working:

Distance to surface: \_\_\_\_\_ m

#### b) Cave (air)

Echo time = 1.20 s  
Speed of sound = 340 m/s

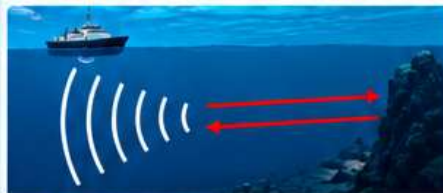


Working:

Distance to surface: \_\_\_\_\_ m

#### c) Underwater sonar

Echo time = 0.80 s  
Speed of sound in water = 1,500 m/s



Working:

Distance to surface: \_\_\_\_\_ m

### 5 FIND THE MISTAKE

A student calculates the distance to a wall. Their working is shown below.

Echo time = 0.60 s  
Speed of sound = 340 m/s  
Distance =  $340 \times 0.60 = 204$  m  
The wall is 204 m away.

a) What mistake has the student made?

\_\_\_\_\_

\_\_\_\_\_

b) What is the correct distance to the wall?

\_\_\_\_\_ m

### 6 CHALLENGE

A bat sends out a sound pulse and receives an echo 0.08 s later. The speed of sound in air is 340 m/s.

a) What is the distance to the object?

\_\_\_\_\_ m

b) If the echo time was 0.16 s, how far away would the object be?

\_\_\_\_\_ m

c) A student calculates a distance of 170 m for an echo time of 1.00 s. Is this correct? Explain.

\_\_\_\_\_

\_\_\_\_\_



### KEY TAKEAWAYS

- ✓ Use distance = speed  $\times$  time.
- ✓ For an echo, the measured time is for the out-and-back journey.
- ✓ Distance to surface =  $\frac{\text{speed} \times \text{time}}{2}$
- ✓ Typical speed of sound: 340 m/s in air, 1,500 m/s in water (varies with conditions).
- ✓ Echoes are used in real life for distance measurement, including sonar and navigation.
- ✓ Always check if you need to divide by two!



3 SOUND

DESIGN CHALLENGE | 50-60 MIN

# DESIGN A BETTER PERFORMANCE SPACE



## LEARNING OBJECTIVE:

Apply reflection, absorption, frequency and sound-wave concepts to acoustic design.

Good sound  
is not an accident.  
It's good physics.

### 1 SCENARIO

You are part of a design team asked to improve a small performance room for live music, such as a school theatre or rehearsal space. The current room has poor acoustics and needs to be redesigned.



**Goal:** Create a space with clear, balanced sound for both performers and the audience.

### 2 CURRENT PROBLEMS

- ! **Excessive echoes**  
(many reflections)
- ! **Uneven sound**  
(some areas too loud, others too quiet)
- ! **External noise**  
(from outside the room)
- ! **Audience audibility**  
(need to hear clearly)

### 3 DESIGN YOUR SOLUTION

Use the room diagram below to plan your design. You can add, remove or modify features.

- Absorbing material**  
(e.g. acoustic panels, curtains)
- Reflective material**  
(e.g. wood, concrete, glass)
- Diffuser**  
(scatter sound)
- Structural features**  
(e.g. angled walls, ceiling panels)

#### PERFORMANCE ROOM (CUTAWAY VIEW)

**Ceiling:**  
What material?  
Why?

**Left wall:**  
What treatment?  
Why?

**Window:**  
How will you reduce  
external noise?

**Back wall:**  
What treatment?  
Why?

**Right wall:**  
What treatment?  
Why?

**Speaker position:**  
Where will you place  
the speakers?  
Why?

### 4 STUDENT TASK

- 1 Choose suitable surface materials for the walls, ceiling and floor.
- 2 Decide on speaker positions.
- 3 Include any additional features (e.g. curtains, diffusers, angled walls).
- 4 Consider how to reduce external noise.
- 5 Use the diagram to show your design and label your choices.

### 5 JUSTIFICATION

For each decision, explain the physics reason. Consider:

- reflection
- absorption
- frequency (high and low)
- diffusion
- how sound spreads in the room
- how your design solves the problems listed above

### 6 CHALLENGE

How can you reduce unwanted reflections without making the room sound "dead"?

Explain how you will balance absorption and reflection to achieve clear, natural sound for both performers and the audience.



### KEY TAKEAWAYS

- ✓ Room design affects sound quality.
- ✓ Different materials absorb or reflect sound.
- ✓ Controlling reflections improves clarity.
- ✓ Good acoustic design balances absorption and useful reflections.
- ✓ Physics is used in real-life performance spaces, studios, theatres and concert halls.



4 LIGHT & OPTICS

CONCEPT LESSON | 50-60 MIN

# LIGHT CARRIES ENERGY AND INFORMATION



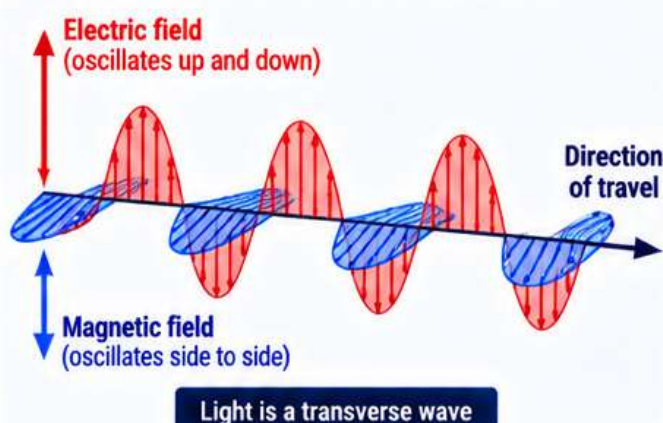
## LEARNING OBJECTIVE:

Describe light as an electromagnetic transverse wave.

Light helps us see the world and carries energy across huge distances.

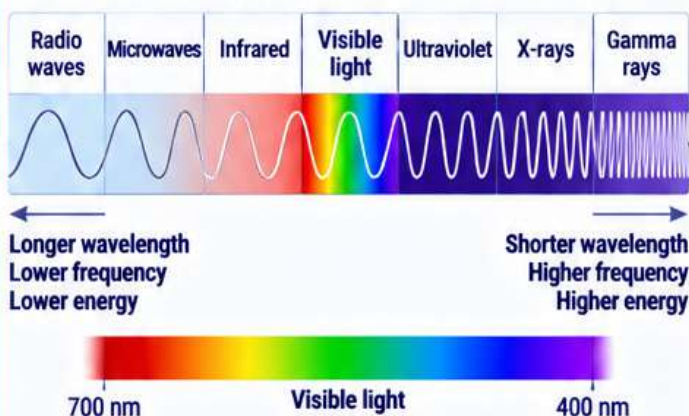
### 1 WHAT IS LIGHT?

Light is electromagnetic radiation. It is a transverse wave made of oscillating electric and magnetic fields. Light can travel through a vacuum, so it does not need a medium.



### 2 THE ELECTROMAGNETIC SPECTRUM

Visible light is only a small part of a much larger range of electromagnetic waves. All electromagnetic waves travel at the same speed in a vacuum ( $3.0 \times 10^8$  m/s).



### 3 COMPARING LIGHT AND SOUND

Use the information below to complete the comparison table.

| Property                         | Light | Sound |
|----------------------------------|-------|-------|
| Wave type                        | _____ | _____ |
| Need for a medium                | _____ | _____ |
| Can travel through a vacuum?     | _____ | _____ |
| Typical speed (in air or vacuum) | _____ | _____ |
| Example of what it carries       | _____ | _____ |



#### DID YOU KNOW?

The Sun is about 150 million kilometres from Earth. Its light takes about 8 minutes to reach us, but its sound cannot travel through the empty space between the Sun and Earth.



### 4 STUDENT TASK

- Complete the comparison table.
- Explain why light is a transverse wave.
- Identify which waves in the electromagnetic spectrum have higher energy than visible light.
- Give one real-world example of how light carries information.

### 5 CHALLENGE

Explain why sunlight can reach Earth while sound from the Sun cannot travel through empty space.

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#### KEY TAKEAWAYS

- Light is electromagnetic radiation.
- Light is a transverse wave.
- Light can travel through a vacuum.
- Visible light is only a small part of the electromagnetic spectrum.
- Light and sound are different types of waves.
- Sunlight can reach Earth because light travels through space, but sound cannot travel without a medium.



4 LIGHT & OPTICS

DIAGRAM LAB ⌚ 50-60 MIN

# THE LAW OF REFLECTION

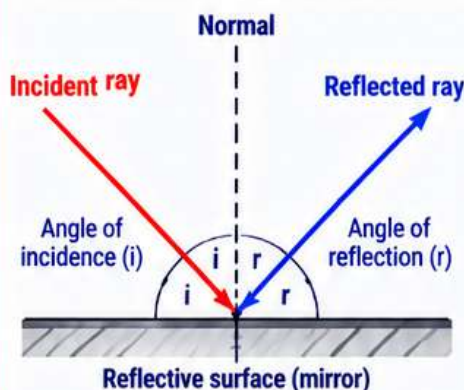
Light reflects in a predictable way. The angle in equals the angle out!



## LEARNING OBJECTIVE:

Apply the law of reflection to light rays.

### 1 KEY VOCABULARY



**Incident ray** – the incoming light ray

**Reflected ray** – the light ray that bounces off the surface

**Normal** – a line at 90° to the surface

**Angle of incidence (i)** – the angle between the incident ray and the normal

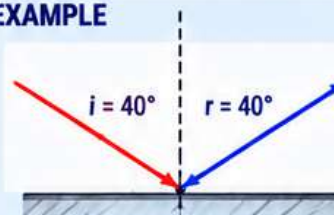
**Angle of reflection (r)** – the angle between the reflected ray and the normal

### 2 THE CORE RULE

**Angle of incidence = Angle of reflection**  
 $i = r$

This means the incident ray, reflected ray and normal all lie in the same plane.

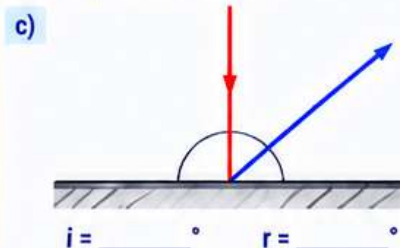
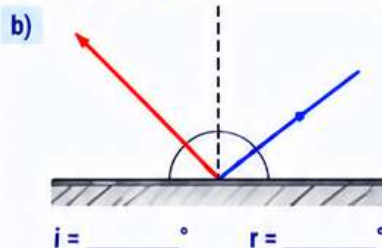
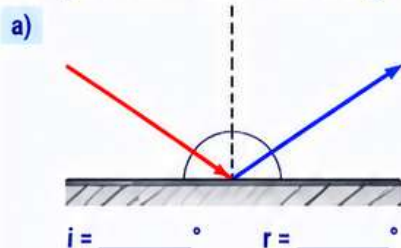
#### EXAMPLE



If the angle of incidence is 40°, the angle of reflection is also 40°.

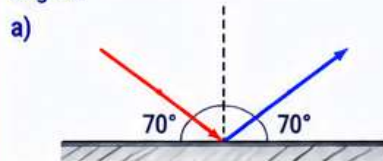
### 3 STUDENT TASK – MEASURE THE ANGLES

Use a protractor to measure the angle of incidence and the angle of reflection in each diagram. Complete the values.



### 4 ERROR ANALYSIS

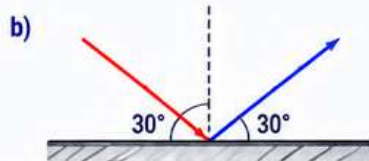
The diagrams below show students who have measured angles from the mirror surface instead of from the normal. Identify the mistake and write the correct angles.



What is the mistake?

Correct angle of incidence: \_\_\_\_\_°

Correct angle of reflection: \_\_\_\_\_°



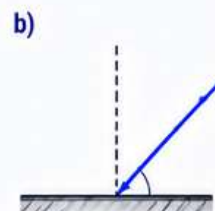
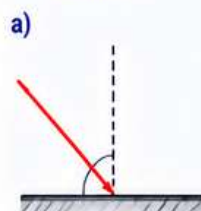
What is the mistake?

Correct angle of incidence: \_\_\_\_\_°

Correct angle of reflection: \_\_\_\_\_°

### 5 CHALLENGE

Before drawing, predict the direction of the reflected ray for each incident ray. Then draw it and check your answer.



### 6 THINK FURTHER

How would the reflected ray change if the surface was tilted? Draw a quick diagram and explain.



#### KEY TAKEAWAYS

- ✓ Angle of incidence = angle of reflection ( $i = r$ ).
- ✓ Angles are measured from the normal, not the surface.
- ✓ The incident ray, reflected ray and normal lie in the same plane.
- ✓ The law of reflection applies to all smooth, reflective surfaces.
- ✓ You can predict the direction of the reflected ray using this rule.



4 LIGHT & OPTICS

VISUAL PROBLEM SOLVING 50-60 MIN

# WHERE DOES THE IMAGE APPEAR?

Same size.  
Upright.  
Virtual.  
But reversed!



## LEARNING OBJECTIVE:

Construct ray diagrams for plane mirrors and describe the resulting image.

### 1 IMAGE PROPERTIES IN A PLANE MIRROR

A plane mirror forms a virtual image with the following properties:

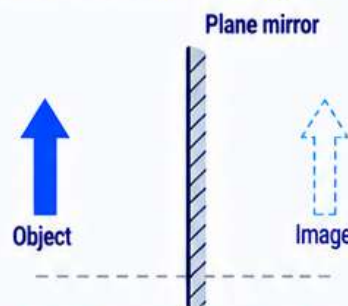
- ✓ **Virtual** (cannot be projected on a screen)
- ✓ **Upright** (same orientation as the object)
- ✓ **Same size** as the object
- ✓ **Laterally reversed** (left-right reversed)
- ✓ **Equal distance** behind the mirror as the object is in front of the mirror



The image is the same distance behind the mirror as the object is in front.

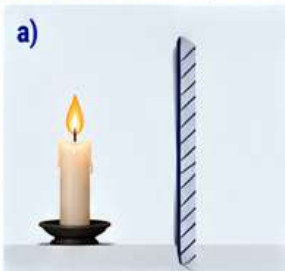
### 2 BASIC RAY DIAGRAM

Complete the ray diagram to locate the image of the arrow in the plane mirror. Draw at least two rays.

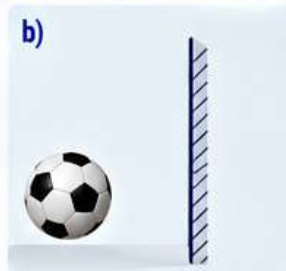


### 3 COMPLETE THE RAY DIAGRAMS

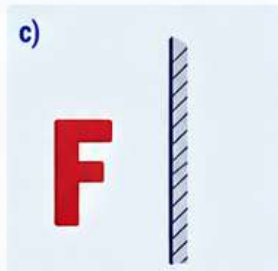
Draw the reflected rays and locate the image in each case.



Locate the image of the candle.



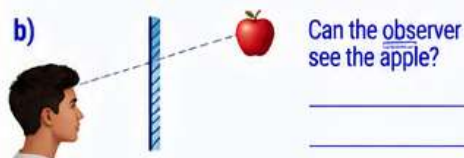
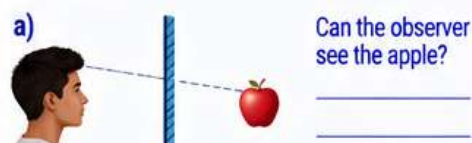
Locate the image of the ball.



Locate the image of the letter.

### 4 CAN THE OBSERVER SEE IT?

In each situation, decide whether the observer can see the object in the mirror. Explain your reasoning.



### 5 REAL-WORLD APPLICATIONS

Plane mirrors are used in many everyday situations.



#### Car mirrors

Help drivers see objects behind them.



#### Bathroom mirrors

Form a clear, upright image.

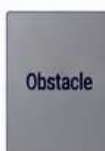


#### Periscopes

Use mirrors to change the direction of light.

### 6 CHALLENGE

Design the minimum mirror arrangement needed to view around the obstacle. Draw your design and explain how it works.



#### KEY TAKEAWAYS

- ✓ A plane mirror forms a virtual, upright, same-size image.
- ✓ The image is laterally reversed (left-right reversed).
- ✓ The image is the same distance behind the mirror as the object is in front.
- ✓ Ray diagrams help us predict where the image appears.
- ✓ You can only see an object if a reflected ray from the object reaches your eye.
- ✓ Mirrors are used in everyday devices to change the direction of light and to help us see more.



4 LIGHT & OPTICS

CONCEPT LESSON | 50-60 MIN

# WHY DOES LIGHT BEND?

Light changes speed when it enters a different material. At an angle, this can change its direction.



## LEARNING OBJECTIVE:

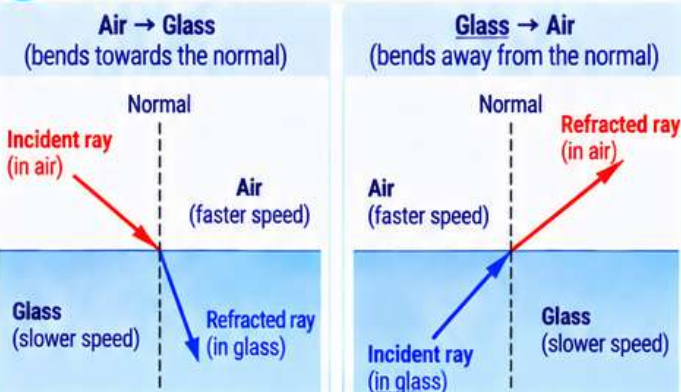
Explain changes in light direction when light crosses boundaries between materials.

### 1 WHAT IS REFRACTION?

Refraction is the change in direction of a light ray when it passes from one material to another because its speed changes.

- Light travels at different speeds in different materials.
- When light crosses a boundary at an angle, its direction can change (bend).
- If the light travels perpendicular to the boundary (straight on), it does not change direction.

### 2 RAY DIAGRAMS



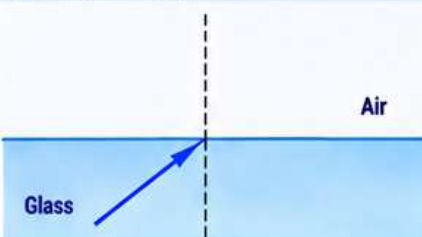
### 3 COMPLETE THE RAY DIAGRAMS

Draw the refracted ray in each diagram. Include the normal and show the correct direction of bending.

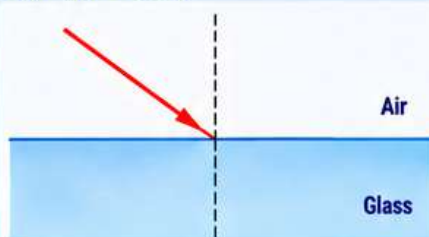
#### a) Air → Water



#### b) Glass → Air



#### c) Air → Glass

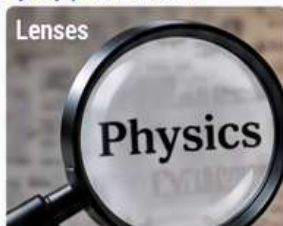


### 4 REAL-WORLD PHYSICS

Refraction explains many everyday phenomena.



Objects under water appear closer to the surface than they really are.



Lenses use refraction to focus light.



Prisms use refraction to separate white light into a spectrum of colours.

### 5 CHALLENGE

Explain why a ray travelling from air into glass bends towards the normal, but a ray travelling from glass into air bends away from the normal. Use the idea of light speed in your explanation.

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### 6 THINK FURTHER

A light ray hits a glass block at different angles.

- 1 What happens to the ray if it hits the boundary at 90°?
- 2 How does the amount of bending change if the ray hits at a larger angle?
- 3 Why might refraction be useful in optical fibres?



### KEY TAKEAWAYS

- ✓ Light changes speed when it moves between materials.
- ✓ At an angle, this change in speed can change the direction of the ray.
- ✓ Light bends towards the normal when it slows down (e.g. air to glass).
- ✓ Light bends away from the normal when it speeds up (e.g. glass to air).
- ✓ Refraction explains apparent depth, lenses, prisms and many optical devices.



4 LIGHT & OPTICS

PHYSICS LAB ⌚ 50–60 MIN

# INVESTIGATING HOW LIGHT BENDS

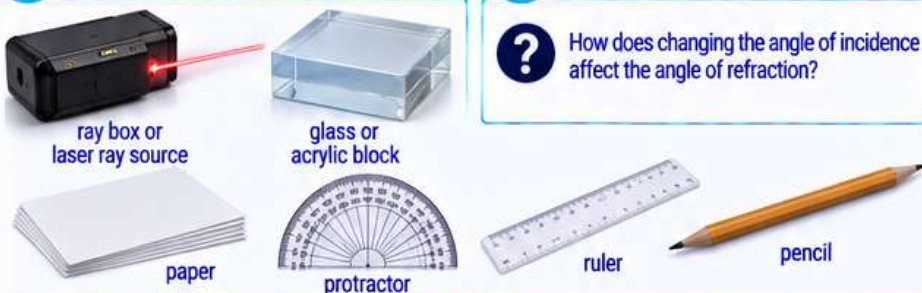
Measure  
Observe  
Think  
Explain!



## LEARNING OBJECTIVE:

Investigate how the angle of incidence affects the direction of a light ray entering a transparent material.

### 1 EQUIPMENT



### 2 INVESTIGATION QUESTION

? How does changing the angle of incidence affect the angle of refraction?

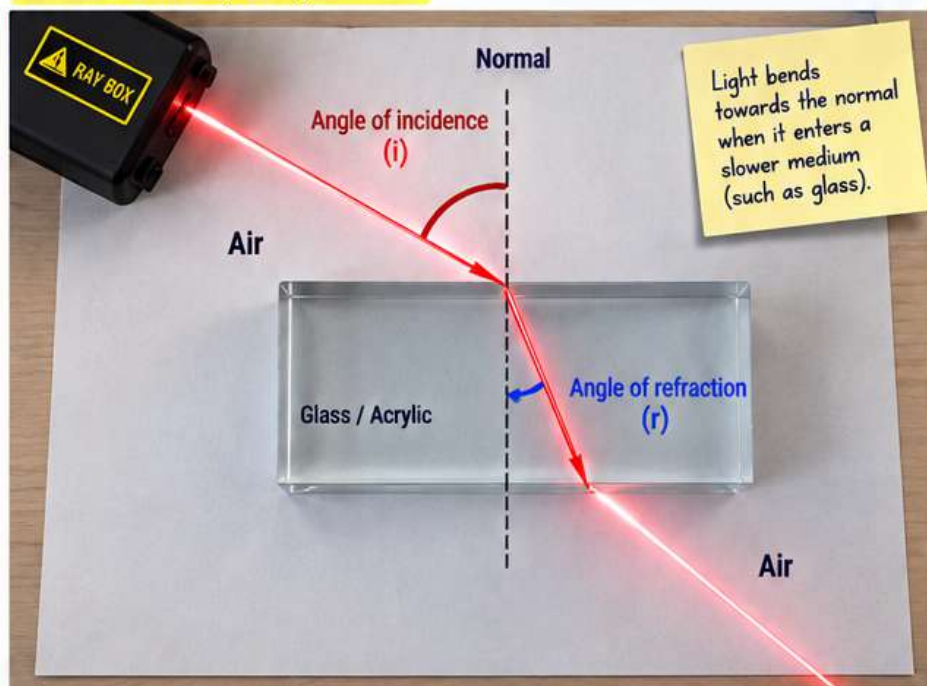
### 3 METHOD

- 1 Place the transparent block on paper and trace its outline.
- 2 Draw a normal at the boundary.
- 3 Direct a light ray toward the block.
- 4 Mark the incident and refracted rays.
- 5 Measure the angle of incidence.
- 6 Measure the angle of refraction.
- 7 Repeat using several incident angles.



**SAFETY:**  
Never direct a light beam toward anyone's eyes.

### 4 EXPERIMENTAL SETUP (TOP VIEW)



### 5 RESULTS TABLE

| Angle of Incidence (°) | Angle of Refraction (°) | Observation |
|------------------------|-------------------------|-------------|
| 10                     |                         |             |
| 20                     |                         |             |
| 30                     |                         |             |
| 40                     |                         |             |
| 50                     |                         |             |

### 6 ANALYSE IT

- 1 What pattern can you identify?
- 2 Does the light bend toward or away from the normal?
- 3 Does the frequency of the light change at the boundary?
- 4 What happens to its speed?

### 7 EVALUATION

Identify:

- 1 one measurement uncertainty
- 2 one possible systematic error
- 3 one improvement

### 8 CHALLENGE

Predict the ray path when light leaves the block and returns to air. Draw the complete path on the diagram below.



### KEY TAKEAWAYS

- ✓ Light changes direction when it crosses a boundary between materials.
- ✓ The amount of bending depends on the angle of incidence.
- ✓ Light bends towards the normal when it enters a slower medium (e.g. air to glass).
- ✓ Light bends away from the normal when it enters a faster medium (e.g. glass to air).
- ✓ The frequency of light does not change at the boundary, but its speed does.
- ✓ Careful measurements help reduce uncertainty and improve results.



4 LIGHT & OPTICS

DIAGRAM LAB ⌚ 50–60 MIN

# CONTROLLING LIGHT WITH LENSES

Same physics.  
Brighter possibilities!

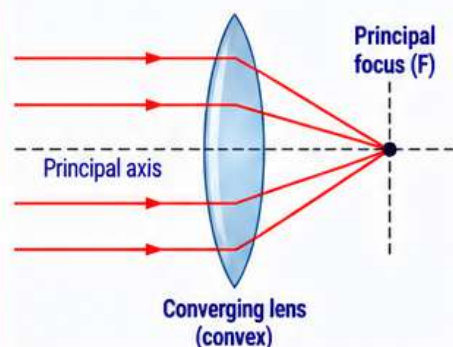


## LEARNING OBJECTIVE:

Explain how converging and diverging lenses change the direction of light rays.

### 1 CONVERGING LENS (CONVEX)

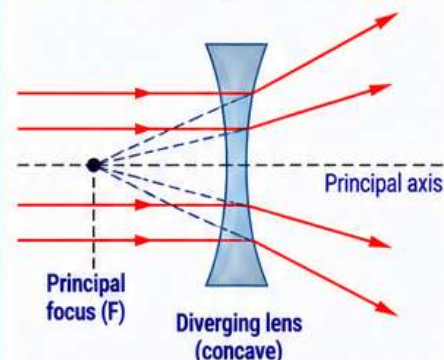
Parallel rays are refracted toward a principal focus.



Rays parallel to the principal axis meet at the principal focus after passing through a converging lens.

### 2 DIVERGING LENS (CONCAVE)

Parallel rays spread apart after refraction.



Rays parallel to the principal axis appear to come from the principal focus after passing through a diverging lens.

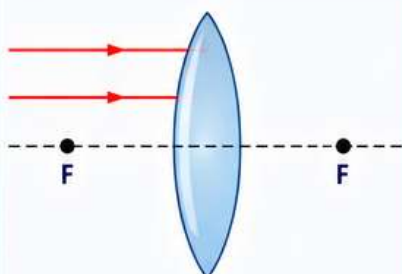
### 3 KEY VOCABULARY

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| <b>Principal axis</b>      | An imaginary straight line through the centre of the lens.                                  |
| <b>Optical centre</b>      | The point at the centre of the lens where light passes through with little or no deviation. |
| <b>Principal focus (F)</b> | The point where parallel rays meet (converging) or appear to come from (diverging).         |
| <b>Focal length (f)</b>    | The distance from the optical centre to the principal focus.                                |
| <b>Converging lens</b>     | A lens that is thicker in the middle and brings light rays together.                        |
| <b>Diverging lens</b>      | A lens that is thinner in the middle and causes light rays to spread apart.                 |

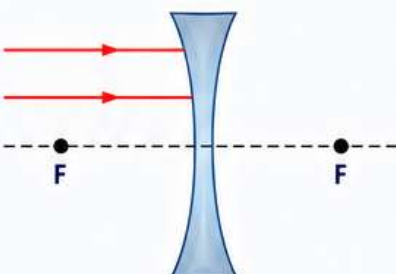
### 4 STUDENT TASK – COMPLETE THE RAY DIAGRAMS

Complete the paths of the rays for each lens.

#### a) Converging lens



#### b) Diverging lens



### 5 REAL-WORLD PHYSICS

Lenses are used in many everyday devices.



**Cameras**  
Focus light to form an image on a sensor.



**Glasses**  
Correct vision by diverging or converging light.



**Telescopes**  
Use lenses to make distant objects appear larger.



**Microscopes**  
Use lenses to magnify small objects.

### 6 PHYSICS THINKING

How does changing the curvature of a lens affect how strongly it bends light?

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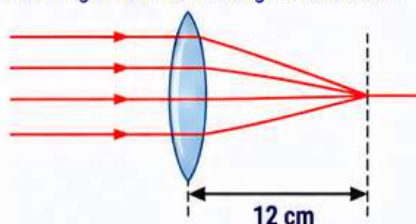
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### 7 CHALLENGE

Determine the focal length of the converging lens in the diagram below. The diagram is to scale.



Focal length (f) = \_\_\_\_\_



### KEY TAKEAWAYS

- ✓ Parallel rays meet at the principal focus in a converging lens.
- ✓ Parallel rays spread apart in a diverging lens.
- ✓ The optical centre is where light passes straight through.
- ✓ The focal length is the distance to the principal focus.
- ✓ Lenses are used in cameras, glasses, telescopes, microscopes and projectors.
- ✓ The curvature of a lens affects how strongly it bends light.



4 LIGHT & OPTICS

VISUAL PROBLEM SOLVING ⌚ 50–60 MIN

# WHERE WILL THE IMAGE FORM?

Same object.  
Different position.  
Different image!

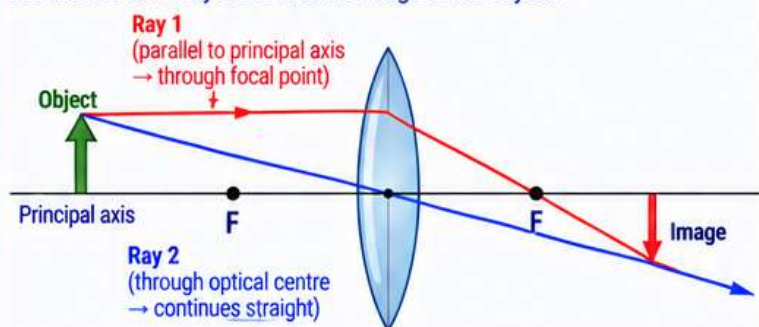


## LEARNING OBJECTIVE:

Use ray diagrams to determine the position and characteristics of images produced by a converging lens.

### 1 RAY DIAGRAM FOR A CONVERGING LENS

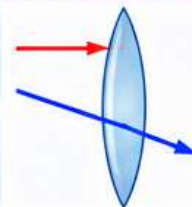
Use the two main rays to locate the image of the object.



The image formed is real, inverted and on the **opposite** side of the lens when the object is **beyond** the focal length.

### 2 CORE RAY RULES

- 1 Parallel to principal axis**  
Refracts through the principal focus on the other side of the lens.
- 2 Through optical centre**  
Continues approximately straight (with little or no deviation).



### 3 IMAGE PROPERTIES

Describe each image in terms of:

- real or virtual
- upright or inverted
- enlarged or diminished
- position (relative to the focal points)

The image type depends on the object distance from the lens.

### 4 STUDENT TASK – COMPLETE THE RAY DIAGRAMS

Draw the missing ray(s) and locate the image. Then describe the image formed.

#### a) Object beyond 2F

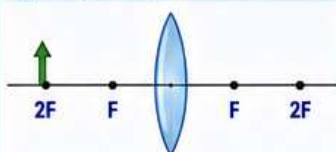


Image position: \_\_\_\_\_  
Real or virtual: \_\_\_\_\_  
Upright or inverted: \_\_\_\_\_  
Enlarged or diminished: \_\_\_\_\_

#### b) Object between F and 2F

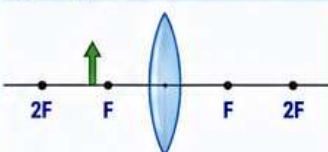


Image position: \_\_\_\_\_  
Real or virtual: \_\_\_\_\_  
Upright or inverted: \_\_\_\_\_  
Enlarged or diminished: \_\_\_\_\_

#### c) Object at F

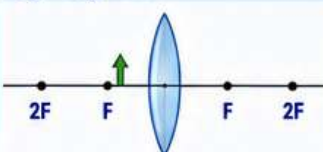


Image position: \_\_\_\_\_  
Real or virtual: \_\_\_\_\_  
Upright or inverted: \_\_\_\_\_  
Enlarged or diminished: \_\_\_\_\_

#### d) Object inside F

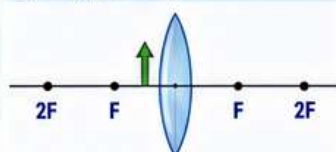


Image position: \_\_\_\_\_  
Real or virtual: \_\_\_\_\_  
Upright or inverted: \_\_\_\_\_  
Enlarged or diminished: \_\_\_\_\_

### 5 REAL-WORLD APPLICATIONS

Converging lenses are used in many devices.



**Cameras**  
Form real, inverted and diminished images on a sensor.



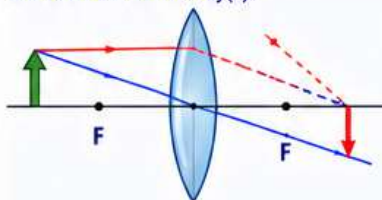
**Magnifying glasses**  
Form virtual, upright and enlarged images (object inside the focal length).



**Projectors**  
Form real, inverted and enlarged images on a screen.

### 6 ERROR ANALYSIS

A student has drawn the ray diagram below. Identify the mistake(s) and draw the correct ray(s).



What is wrong?

Draw the corrected ray diagram.

### 7 CHALLENGE

Explain why moving an object toward a lens can dramatically change the image formed. Your explanation should include ideas about focal length, ray direction and whether the image is real or virtual.



#### KEY TAKEAWAYS

- Use two key rays: parallel to axis (through F) and through optical centre.
- Image properties depend on the object distance.
- Objects beyond F → real, inverted images.
- Objects inside F → virtual, upright, enlarged images.
- Converging lenses are used in cameras, magnifying glasses and projectors.
- Small changes in object position can lead to large changes in image type and size.



4 LIGHT & OPTICS

CONCEPT + VISUAL INVESTIGATION | 50–60 MIN

# WHY DO WE SEE COLOUR?



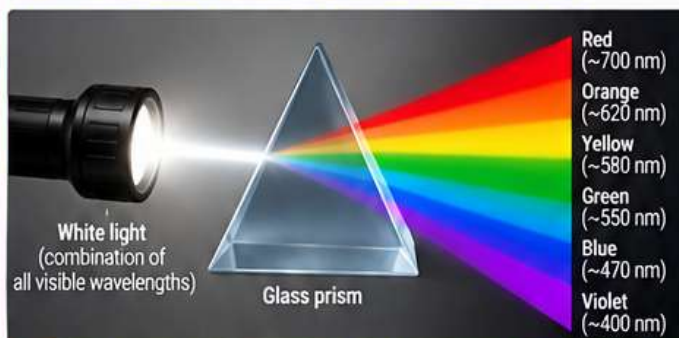
## LEARNING OBJECTIVE:

Explain white light, the visible spectrum and the appearance of coloured objects.

White light  
many colours.  
Objects reflect  
some, absorb others!

### 1 WHITE LIGHT AND THE VISIBLE SPECTRUM

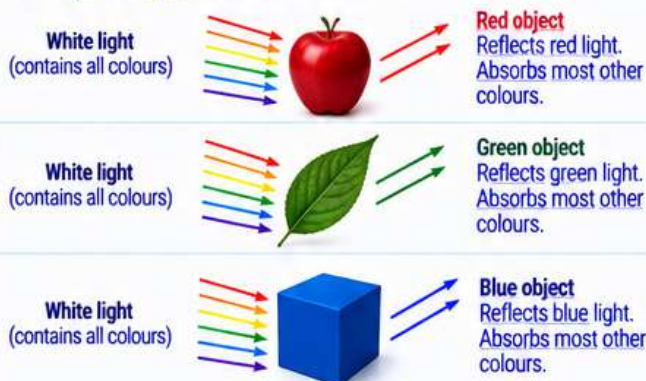
White light contains a range of wavelengths. When it passes through a prism, it is separated into its component colours (dispersion).



The visible spectrum is the range of wavelengths of light that can be seen by the human eye (approximately 400 nm to 700 nm).

### 2 WHY DO OBJECTS APPEAR COLOURED?

An object appears a particular colour because it reflects certain wavelengths of light and absorbs the others.



### 3 COLOURED OBJECTS UNDER DIFFERENT ILLUMINATION

Predict the appearance of each object under the different lights. Write the colour (or if it appears dark).

| Object | White light | Red light | Blue light |
|--------|-------------|-----------|------------|
|        | _____       | _____     | _____      |
|        | _____       | _____     | _____      |
|        | _____       | _____     | _____      |
|        | _____       | _____     | _____      |

### 4 CORE IDEA

- ✓ Visible light contains a range of wavelengths perceived as different colours.
- ✓ An object appears a particular colour because of the wavelengths it reflects or transmits to the observer.
- ✓ White light can be separated into its component colours by dispersion (e.g. using a prism).

### 5 COMMON MISCONCEPTION

**✗** Objects do not "contain" the colour that we see.

| Reality                                                         | Misconception                              |
|-----------------------------------------------------------------|--------------------------------------------|
| A red object reflects red light and absorbs most other colours. | A red object contains red light inside it. |

### 6 REAL-WORLD PHYSICS

Colour is important in many technologies.



**Displays**  
Use coloured light to create images.



**Stage lighting**  
Uses coloured lights to create effects.



**Photography**  
Uses filters and spectrum control.



**Coloured filters**  
Transmit specific colours of light.

### 7 CHALLENGE

Explain why a red object may appear very dark under blue illumination. Use ideas about reflection, absorption and the wavelengths of light.

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### 8 THINK FURTHER

- 1 Why does a yellow object appear yellow instead of red and green?
- 2 How might the colours we see change under different light sources, such as LED lighting?
- 3 Research an example where colour is used for safety (e.g. traffic lights or safety signs). Explain why those colours are effective.



## KEY TAKEAWAYS

- ✓ White light contains a range of wavelengths (400–700 nm).
- ✓ Objects appear the colour they reflect.
- ✓ A prism can separate white light into its component colours.
- ✓ The visible spectrum is: red, orange, yellow, green, blue, violet.
- ✓ Objects do not contain the colour we see.
- ✓ Colour has many real-world applications.



4 LIGHT & OPTICS

CONCEPT LESSON

50-60 MIN

# THE ELECTROMAGNETIC SPECTRUM

Different wavelengths.  
Same speed.  
A whole spectrum of possibilities!

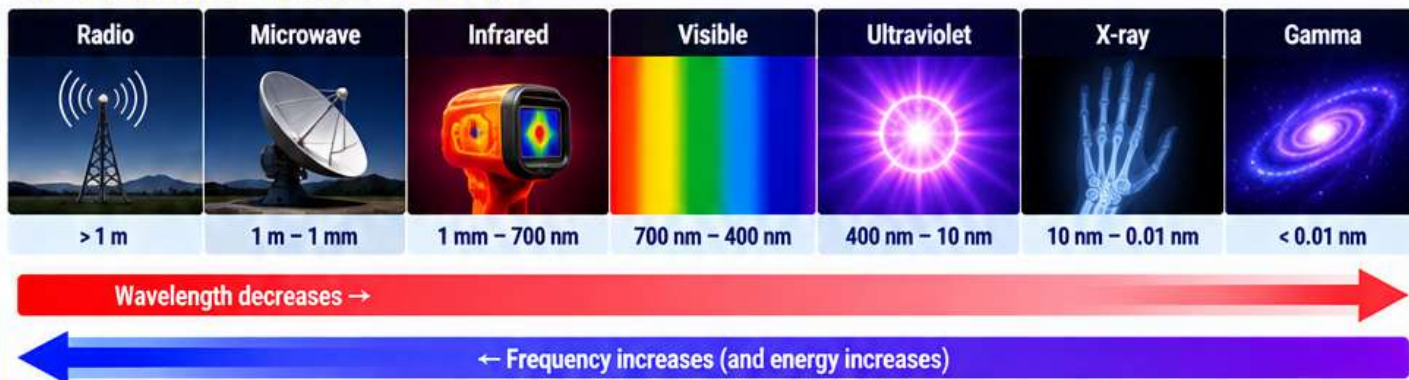


## LEARNING OBJECTIVE:

Describe the electromagnetic spectrum and relate frequency, wavelength and energy across it.

### 1 ONE FAMILY OF WAVES

The electromagnetic spectrum is the complete range of electromagnetic waves, ordered by their wavelength, frequency and energy. All electromagnetic waves have the same speed in a vacuum:  $3.0 \times 10^8$  m/s.



### 2 KEY FACTS ABOUT ELECTROMAGNETIC WAVES

- ✓ All electromagnetic waves are transverse.
- ✓ They can travel through a vacuum (no medium required).
- ✓ They transfer energy.
- ✓ They all travel at the same speed in a vacuum:  $3.0 \times 10^8$  m/s.

### 3 THE WAVE EQUATION

$$v = f\lambda$$

In a vacuum,  $v$  is constant ( $3.0 \times 10^8$  m/s), so if frequency increases, wavelength must decrease, and vice versa.

This is why the electromagnetic spectrum has shorter wavelengths and higher frequencies towards the right.

### 4 COMPLETE THE SPECTRUM

Use the labels below to complete the electromagnetic spectrum.



### 5 COMPARE AND IDENTIFY

Use the spectrum above to answer the questions.

- Which type of wave has the longest wavelength?
- Which type of wave has the shortest wavelength?
- Which type of wave has the lowest frequency?
- Which type of wave has the highest frequency?
- Where is visible light located on the spectrum?

### 6 REAL-WORLD APPLICATIONS

Electromagnetic waves are used in many technologies.



**Radio waves**  
Communication (broadcast, mobile)



**Microwaves**  
Cooking, satellite communication



**Infrared**  
Thermal imaging, night vision



**X-rays**  
Medical imaging, security scanners

### 7 CHALLENGE

Explain why visible light represents only a small portion of the entire electromagnetic spectrum.

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### 8 THINK FURTHER

- Why do all electromagnetic waves travel at the same speed in a vacuum?
- If visible light has a higher frequency than infrared, what can you say about its wavelength?
- How might the uses of electromagnetic waves be different if their speeds in a vacuum were not the same?



## KEY TAKEAWAYS

- ✓ Electromagnetic waves form a continuous spectrum.
- ✓ Frequency increases and wavelength decreases from radio to gamma rays.
- ✓ All electromagnetic waves travel at  $3.0 \times 10^8$  m/s in a vacuum.
- ✓ They are transverse waves that transfer energy.
- ✓ Visible light is only a small part of the electromagnetic spectrum.
- ✓ Different parts of the spectrum have different real-world applications.



4 LIGHT & OPTICS

REAL-WORLD PHYSICS ⌚ 50–60 MIN

# DIFFERENT WAVES. DIFFERENT JOBS.

Same physics.  
Real solutions.  
A brighter world!



## LEARNING OBJECTIVE:

Relate electromagnetic-wave properties to their technological and practical uses.

### 1 THE ELECTROMAGNETIC SPECTRUM AND ITS USES

Each region of the electromagnetic spectrum has unique properties that make it useful for different applications.

| RADIO                                                                                 | MICROWAVE                                                                                  | INFRARED                                                                                   | VISIBLE LIGHT                                                                                                | ULTRAVIOLET                                                                              | X-RAYS                                                            | GAMMA                                                                                         |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
|                                                                                       |                                                                                            |                                                                                            |                                                                                                              |                                                                                          |                                                                   |                                                                                               |
| <ul style="list-style-type: none"> <li>broadcasting</li> <li>communication</li> </ul> | <ul style="list-style-type: none"> <li>satellite communication</li> <li>cooking</li> </ul> | <ul style="list-style-type: none"> <li>thermal imaging</li> <li>remote controls</li> </ul> | <ul style="list-style-type: none"> <li>vision</li> <li>photography</li> <li>optical communication</li> </ul> | <ul style="list-style-type: none"> <li>security marking</li> <li>fluorescence</li> </ul> | <ul style="list-style-type: none"> <li>medical imaging</li> </ul> | <ul style="list-style-type: none"> <li>sterilisation</li> <li>medical applications</li> </ul> |
| > 1 m                                                                                 | 1 m – 1 mm                                                                                 | 1 mm – 700 nm                                                                              | 700 nm – 400 nm                                                                                              | 400 nm – 10 nm                                                                           | 10 nm – 0.01 nm                                                   | < 0.01 nm                                                                                     |

Longer wavelength  
Lower frequency, lower energy

Shorter wavelength  
Higher frequency, higher energy

### 2 STUDENT TASK – MATCH THE APPLICATION

Match each application to the correct region of the electromagnetic spectrum. Write the letter of the region (A–G).

- Cooking food in a microwave oven
- Heating objects in thermal cameras
- Broadcasting TV and radio signals
- Medical imaging of bones
- Security markings on bank notes
- Sterilising medical equipment
- Seeing the world around us
- Satellite communication
- Controlling a TV with a remote
- Optical fibre communication

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- A** Radio  
**B** Microwave  
**C** Infrared  
**D** Visible light  
**E** Ultraviolet  
**F** X-rays  
**G** Gamma

### 5 THINK FURTHER

Explain why selecting an electromagnetic wave for a particular application depends on its properties, not simply its ability to carry energy.

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### 3 EXPLAIN IT

Complete the table for the following applications.

| Wave        | Relevant Property                                        | Application      |
|-------------|----------------------------------------------------------|------------------|
| Microwave   | Penetrates materials and is absorbed by water molecules. | Cooking food     |
| Infrared    | _____                                                    | Thermal imaging  |
| Ultraviolet | _____                                                    | Security marking |
| X-rays      | _____                                                    | Medical imaging  |

### 4 REAL-WORLD CHALLENGE

Choose the most appropriate electromagnetic wave for each problem. Write the name of the region and explain your choice.

| Problem                                               | Electromagnetic wave | Reason |
|-------------------------------------------------------|----------------------|--------|
| A. Communicate with a spacecraft millions of km away. | _____                | _____  |
| B. Detect a person in the dark.                       | _____                | _____  |
| C. Kill bacteria on medical equipment.                | _____                | _____  |
| D. Take an image of a broken bone.                    | _____                | _____  |



### KEY TAKEAWAYS

- ✓ Different electromagnetic waves have different properties.
- ✓ Their properties make them suitable for different applications.
- ✓ The same physics is used in many everyday technologies.
- ✓ Choosing the right wave depends on its interaction with matter, not just its ability to carry energy.



4 LIGHT & OPTICS

EVIDENCE & RISK ANALYSIS ⌚ 50-60 MIN

# BENEFIT VS RISK



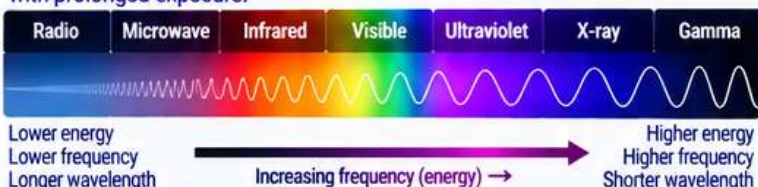
## LEARNING OBJECTIVE:

Evaluate potential biological effects of electromagnetic radiation and explain how exposure can be controlled.

Same physics.  
Real-world  
impacts.  
Informed choices  
for a safer future.

### 1 THE BIG PICTURE

The electromagnetic spectrum provides many benefits, but some types of radiation can be harmful, especially at high energies or with prolonged exposure.



### 2 CORE PHYSICS

Different parts of the electromagnetic spectrum interact with matter differently. The potential effects depend on:

- ✓ frequency (energy of the photons)
- ✓ intensity (amount of radiation)
- ✓ exposure duration (time)
- ✓ type of material (e.g. living tissue)

Higher-energy radiation (UV, X-rays and gamma rays) can ionise atoms and molecules, which may damage cells and DNA.

### 3 EXAMPLES: USES AND RISKS

#### ULTRAVIOLET (UV)



##### Useful applications:

- sterilisation (e.g. UV lamps)
- security markings
- tanning (controlled exposure)

##### Potential risks:

- can damage skin and eyes
- can increase the risk of skin cancer



#### X-RAYS



##### Useful applications:

- medical imaging (bones, teeth)
- security scanning (airports)
- industrial inspection

##### Potential risks:

- ionising radiation capable of damaging cells and DNA



#### GAMMA RAYS



##### Useful applications:

- cancer treatment (radiotherapy)
- sterilisation of medical equipment
- industrial inspection

##### Potential risks:

- highly penetrating ionising radiation
- can damage cells and DNA



### 4 STUDENT TASK – COMPLETE THE TABLE

Use the information on this page to complete the table.

| Radiation | Useful Application | Potential Risk | Risk Reduction |
|-----------|--------------------|----------------|----------------|
| UV        |                    |                |                |
| X-ray     |                    |                |                |
| Gamma     |                    |                |                |

### 5 RISK REDUCTION STRATEGIES

The risks from electromagnetic radiation can be reduced by:



##### Shielding

Use materials such as lead or concrete (for X-rays and gamma rays).



**Reducing exposure time**  
Limit the time of exposure.



**Increasing distance**  
Intensity decreases with distance.



**Protective materials**  
Use appropriate eye protection, clothing or filters (e.g. UV goggles).

### 6 PHYSICS THINKING



Why does "radiation" not automatically mean "dangerous"?

### 7 CHALLENGE

#### CHALLENGE

Evaluate a real-world situation where the benefit of controlled radiation exposure may outweigh the risk. Explain your reasoning.



#### KEY TAKEAWAYS

- ✓ Different parts of the spectrum interact with matter differently.
- ✓ Potential effects depend on frequency, intensity, exposure time and the material.
- ✓ Many applications of electromagnetic radiation provide significant benefits.
- ✓ Risks can be reduced using shielding, less time, greater distance and protective materials.
- ✓ It is possible to use electromagnetic radiation safely when risks are managed.



#### REAL-WORLD IMPACT

Understanding the risks and benefits of electromagnetic radiation helps us make informed decisions in medicine, industry, communication and everyday life.

Science helps us use powerful ideas safely.



4 LIGHT & OPTICS

APPLIED PHYSICS 50-60 MIN

# HOW DOES INFORMATION TRAVEL?

Waves connect  
our world.  
Information  
travels fast!



## LEARNING OBJECTIVE:

Explain how different electromagnetic waves are used to transmit information.

### 1 COMMUNICATION SYSTEMS

Electromagnetic waves are used in many technologies to transmit information.



**Radio broadcasting**  
Uses radio waves to transmit audio signals.



**Mobile communication**  
Uses microwaves (radio waves) for voice and data.



**Satellites**  
Use microwaves for communication over long distances.



**Wi-Fi**  
Uses radio/microwave waves to transmit data wirelessly.



**Fibre-optic systems**  
Uses visible or infrared light to transmit data through optical fibres.

### 2 THE COMMUNICATION MODEL

Any communication system follows the same basic model:



The information is carried by a wave, but the wave itself is changed (modulated) to encode the message.

### 3 A CONNECTED WORLD

Different communication technologies work together to connect people around the world.



#### SAME PHYSICS. A MORE CONNECTED WORLD.

From local radio to global satellite networks, electromagnetic waves make modern communication possible.

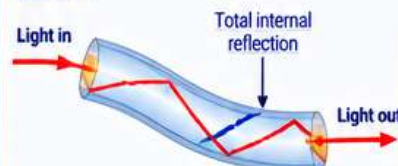
### 4 STUDENT TASK – MATCH THE TECHNOLOGY

Match each communication requirement to the most appropriate technology.

| Requirement                                    | Technology (A-E)         |                                    |
|------------------------------------------------|--------------------------|------------------------------------|
| 1. Broadcast music to many people in a city.   | <input type="checkbox"/> | A Radio waves (broadcasting)       |
| 2. Provide internet to a remote rural area.    | <input type="checkbox"/> | B Microwaves (satellite)           |
| 3. Fast internet connection within a building. | <input type="checkbox"/> | C Wi-Fi (radio/microwave)          |
| 4. Transmit data across an ocean.              | <input type="checkbox"/> | D Fibre optics (optical)           |
| 5. High-speed data with low signal loss.       | <input type="checkbox"/> | E Mobile network (radio/microwave) |

### 5 FIBRE OPTICS – HOW IT WORKS

Light can travel long distances through an optical fibre by total internal reflection.



The light reflects repeatedly inside the fibre, allowing it to travel long distances with very low loss.

### 6 SATELLITE COMMUNICATION

Satellites receive signals from a ground station, amplify them and send them back to Earth.



### 7 COMPARE – WIRED VS WIRELESS

Complete the table to compare optical fibre communication with wireless communication (e.g. radio, microwave or satellite).

| Feature       | Optical Fibre (Wired) | Wireless (Radio/Microwave/Satellite) |
|---------------|-----------------------|--------------------------------------|
| Wave type     |                       |                                      |
| Typical speed |                       |                                      |
| Signal loss   |                       |                                      |
| Mobility      |                       |                                      |
| Common uses   |                       |                                      |

### 8 CHALLENGE

A remote research station needs long-distance communication. Select an appropriate system and justify the decision using wave properties (e.g. wavelength, ability to travel long distances, need for line of sight).

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#### KEY TAKEAWAYS

- ✓ Different electromagnetic waves are used for different communication systems.
- ✓ Information is encoded onto a wave (modulation).
- ✓ Fibre optics use light and total internal reflection.
- ✓ Satellites use microwaves to communicate over long distances.
- ✓ The best technology depends on the situation and the wave properties.



#### REAL-WORLD IMPACT

Electromagnetic waves help people communicate across cities, countries and even space, connecting families, businesses and emergency services around the world.



4 LIGHT & OPTICS

DATA DETECTIVE ⌚ 50-60 MIN

# DECODE THE WAVE DATA

Real data.  
Real thinking.  
Real physics.



## LEARNING OBJECTIVE:

Analyse quantitative wave data and connect frequency, wavelength, speed and wave behaviour.

### 1 THE DATASET

The table shows six unidentified waves (A-F). Each wave has one missing value (—) and one anomalous measurement (which does not fit the expected pattern).

| Wave | Frequency $f$ (Hz)   | Wavelength $\lambda$ (m) | Speed $v$ (m/s)   | Medium | Distance (m) | Travel time $t$ (s)  |
|------|----------------------|--------------------------|-------------------|--------|--------------|----------------------|
| A    | $5.0 \times 10^6$    | 60                       | —                 | Air    | 3 000        | $6.0 \times 10^{-3}$ |
| B    | $2.4 \times 10^9$    | 0.125                    | —                 | Air    | 750          | $2.5 \times 10^{-6}$ |
| C    | $4.0 \times 10^{14}$ | 750                      | $3.0 \times 10^8$ | Vacuum | —            | $5.0 \times 10^{-3}$ |
| D    | $6.0 \times 10^{14}$ | 500                      | $3.0 \times 10^8$ | Vacuum | 1 500        | $5.0 \times 10^{-6}$ |
| E    | $3.0 \times 10^{19}$ | 0.010                    | —                 | Vacuum | 300          | $1.0 \times 10^{-6}$ |
| F    | $2.0 \times 10^{18}$ | 0.20                     | $3.0 \times 10^8$ | Vacuum | 6 000        | $1.0 \times 10^{-5}$ |

### 2 MISSION 1 – CALCULATE

Use the wave equation to calculate the missing wave speeds.

$$v = f\lambda$$

Wave A:  $v =$  \_\_\_\_\_ m/s

Wave B:  $v =$  \_\_\_\_\_ m/s

Wave E:  $v =$  \_\_\_\_\_ m/s

### 3 MISSION 2 – IDENTIFY

Which waves have measurements consistent with travelling in the same medium?

- ☐ A and B      ☐ E and D  
☐ C and D      ☐ B and E  
☐ E and F      ☐ D and F

Explain your reasoning based on the values.

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### 4 MISSION 3 – FIND THE ANOMALY

One of the measurements in the table is inconsistent with the overall data. Which one is it?

- ☐ Wave A      ☐ Wave D  
☐ Wave B      ☐ Wave E  
☐ Wave C      ☐ Wave F

Explain why this measurement is anomalous.

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### 5 MISSION 4 – COMPARE

Rank the waves from highest to lowest for each quantity. Write the letters (A-F) in order.

1. Frequency (highest → lowest): \_\_\_\_\_

2. Wavelength (longest → shortest): \_\_\_\_\_

3. Period (longest → shortest): \_\_\_\_\_

### 6 MISSION 5 – INTERPRET

A wave crosses from one medium into another and slows down. Predict what happens to:

a) The wavelength:

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b) The frequency:

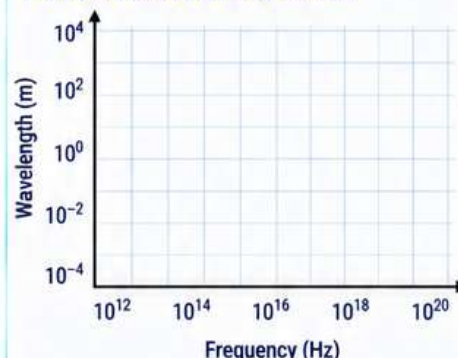
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### 7 MISSION 6 – GRAPH

Construct a graph using the data for waves C, D and F. Plot wavelength (m) on the y-axis and frequency (Hz) on the x-axis (use a log scale if needed).



### 8 CHALLENGE

Use your graph to estimate an unmeasured wavelength or frequency for a wave in the same region of the spectrum. Explain your method and reasoning.

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#### KEY TAKEAWAYS

- ✓ Waves follow the equation  $v = f\lambda$ .
- ✓ Frequency, wavelength and speed are linked.
- ✓ Waves can be identified using their properties and the medium.
- ✓ Data analysis helps detect errors and understand wave behaviour.



#### REAL-WORLD CONNECTION

Scientists and engineers analyse wave data to design communication systems, medical imaging, satellite links and many other technologies.



4 LIGHT & OPTICS

INTEGRATED PROBLEM SOLVING 50–60 MIN

# THE COASTAL COMMUNICATION CHALLENGE

Real problems.  
Multiple concepts.  
Real solutions.



## LEARNING OBJECTIVE:

Combine wave, sound, light and electromagnetic concepts to solve a multi-stage real-world problem.

### 1 SCENARIO

A coastal research facility needs systems for:

- underwater distance measurement
- visual observation
- wireless communication
- optical data transmission
- hazard detection

Students receive:

- wave frequencies
- wavelengths
- sound speeds
- echo times
- optical diagrams
- electromagnetic-wave information



### 2 MISSION 1 – WAVE SPEED

A wave has a frequency of  $2.5 \times 10^6$  Hz and a wavelength of 0.60 m. Calculate the wave speed.

$$v = f\lambda$$

$v =$  \_\_\_\_\_ m/s

### 3 MISSION 2 – UNDERWATER SOUND

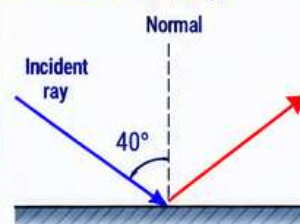
A sonar pulse is sent vertically downward in seawater (speed of sound = 1 500 m/s). The echo returns after 4.0 s. Calculate the distance to the underwater object.



Distance = \_\_\_\_\_ m

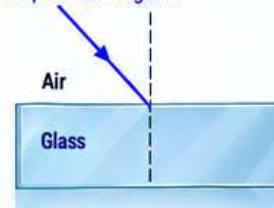
### 4 MISSION 3 – REFLECTION

Complete the diagram to show the reflected wave (or ray).



### 5 MISSION 4 – REFRACTION

A light ray in air enters a transparent observation window (glass). Predict how the ray changes direction. Complete the diagram.



### 6 MISSION 5 – COMMUNICATION

Select the most appropriate electromagnetic wave for long-distance communication with another research station. Justify your choice.

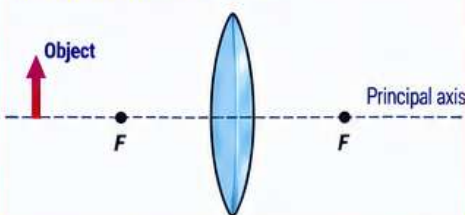
- |                                      |                                        |
|--------------------------------------|----------------------------------------|
| <input type="checkbox"/> Radio waves | <input type="checkbox"/> Visible light |
| <input type="checkbox"/> Microwaves  | <input type="checkbox"/> Ultraviolet   |
| <input type="checkbox"/> Infrared    | <input type="checkbox"/> X-rays        |
|                                      | <input type="checkbox"/> Gamma rays    |

Reason:

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

### 7 MISSION 6 – OPTICAL SYSTEM

Complete the ray diagram for a converging lens used in an observation device.



### 8 MISSION 7 – RISK

Identify one electromagnetic-radiation risk associated with equipment at the facility and propose an appropriate control.

Equipment:

\_\_\_\_\_

Potential risk:

\_\_\_\_\_

Control measure:

\_\_\_\_\_

### 9 FINAL SYSTEM DECISION

Recommend the best combination of technologies for the coastal research facility. Each recommendation must contain the technology, wave type, relevant property and reason.

| System / Purpose          | Technology | Wave Type | Relevant Property | Reason |
|---------------------------|------------|-----------|-------------------|--------|
| Underwater measurement    | _____      | _____     | _____             | _____  |
| Visual observation        | _____      | _____     | _____             | _____  |
| Wireless communication    | _____      | _____     | _____             | _____  |
| Optical data transmission | _____      | _____     | _____             | _____  |
| Hazard detection          | _____      | _____     | _____             | _____  |

### 10 CHALLENGE

One system must continue functioning in darkness and poor weather. Evaluate which wave-based technology is most appropriate. Justify your answer using wave properties.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_



## KEY TAKEAWAYS

- ✓ Different wave types are used for different real-world applications.
- ✓ Wave properties (frequency, wavelength, speed) determine their suitability.
- ✓ Combining physics concepts helps solve complex, real-life problems.
- ✓ Careful analysis leads to better technological decisions.



## REAL-WORLD IMPACT

Wave-based technologies help coastal research, improve safety, support communication and allow us to explore and protect our oceans.



4 LIGHT & OPTICS

UNIT ASSESSMENT 50-60 MIN

# CHECKPOINT – WHAT CAN YOU DO?

Same knowledge.  
Bigger possibilities.



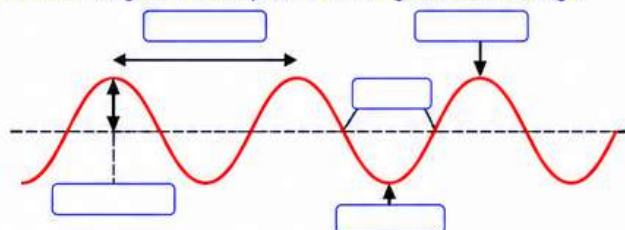
## LEARNING OBJECTIVE:

Demonstrate independent understanding of the major wave, sound, light and electromagnetic skills developed throughout Volume 4.

### A WAVE FUNDAMENTALS



- 1 Explain how waves transfer energy without overall transfer of matter.
- 2 Distinguish between transverse and longitudinal waves.
- 3 Label the diagram with: amplitude, wavelength, crest and trough.

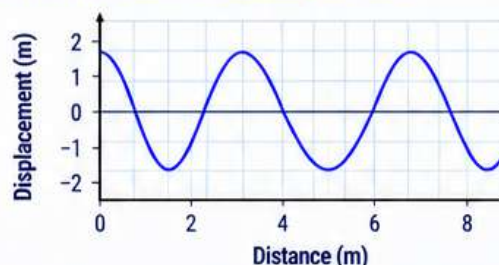


- 4 A wave has a frequency of 8 Hz. Calculate its period.
- 5 A wave has frequency 12 Hz and wavelength 2.5 m. Calculate its speed.

### B WAVE BEHAVIOUR



- 6 Describe reflection.
- 7 Explain why refraction can occur when a wave enters another medium.
- 8 Identify the conditions that produce noticeable diffraction.
- 9 Interpret the displacement–distance graph below.



### C SOUND



- 10 Explain how sound travels through air.
- 11 State how frequency affects pitch.
- 12 State how amplitude relates to sound intensity and perceived loudness.
- 13 Explain why sound cannot travel through a vacuum.
- 14 An echo returns after 0.60 s. Use a sound speed of 340 m/s to calculate the distance to the reflecting surface.



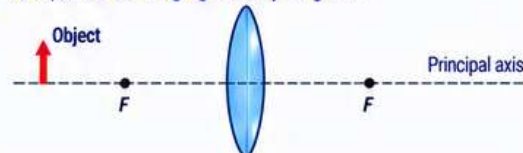
### D LIGHT



- 15 Apply the law of reflection to complete the ray diagram.



- 16 Explain why light changes direction during refraction.
- 17 Complete a converging-lens ray diagram.



- 18 Explain why a coloured object can appear different under coloured illumination.

### E ELECTROMAGNETIC WAVES



- 19 Put the seven regions of the electromagnetic spectrum in order (from lowest to highest frequency).

□ → □ → □ → □ → □ → □ → □

- 20 Select an appropriate electromagnetic wave for each application.

| Application             | Wave type |
|-------------------------|-----------|
| Satellite communication | _____     |
| Thermal imaging         | _____     |
| Medical imaging         | _____     |

- 21 Explain one risk associated with ionising electromagnetic radiation.

### F SCIENTIFIC REASONING



- 22 A student measures the wavelength from a diagram several times and obtains the following results:

0.42 m    0.41 m    0.68 m    0.42 m

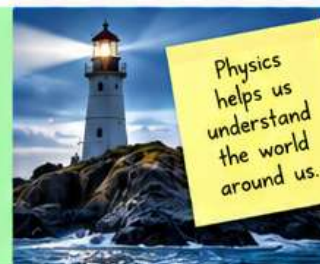
Identify the anomalous result and explain what should happen next.

### SELF-CHECK – CAN YOU DO IT?

- |                                                                |                                                                       |
|----------------------------------------------------------------|-----------------------------------------------------------------------|
| <input type="checkbox"/> I can identify wave properties.       | <input type="checkbox"/> I can analyse sound waves.                   |
| <input type="checkbox"/> I can calculate frequency and period. | <input type="checkbox"/> I can use echo calculations.                 |
| <input type="checkbox"/> I can use the wave equation.          | <input type="checkbox"/> I can construct ray diagrams.                |
| <input type="checkbox"/> I can explain reflection.             | <input type="checkbox"/> I can explain lenses.                        |
| <input type="checkbox"/> I can explain refraction.             | <input type="checkbox"/> I can describe the electromagnetic spectrum. |
| <input type="checkbox"/> I can explain diffraction.            | <input type="checkbox"/> I can evaluate wave applications and risks.  |

### REAL-WORLD SKILLS

- ✓ Apply physics to real situations.
- ✓ Analyse data and identify errors.
- ✓ Use diagrams and calculations.
- ✓ Make informed decisions about technology.



Physics helps us understand the world around us.



4 LIGHT & OPTICS

FINAL PERFORMANCE CHALLENGE ⌚ 60–90 MIN

# DESIGN THE SMART RESCUE NETWORK

Real physics.  
Real problems.  
Real solutions.



## LEARNING OBJECTIVE:

Apply the complete set of Volume 4 skills to design and evaluate a multi-wave communication, detection and imaging system.

## THE SCENARIO

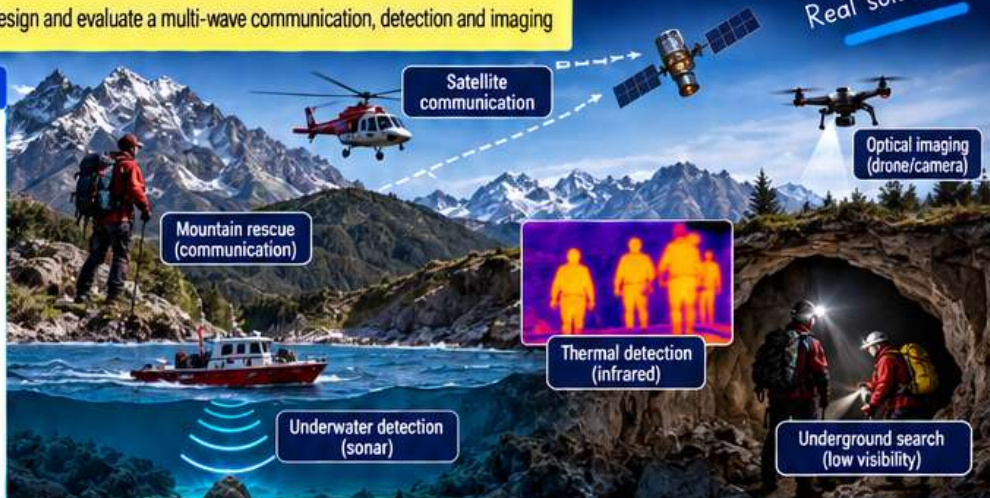
A rescue team must operate across mountains, coastal water, underground structures and low-visibility conditions.

The team needs:

- long-distance communication
- underwater detection
- thermal detection
- optical imaging
- accurate distance measurements

Students receive:

- terrain map
- wave-property data
- frequency/wavelength values
- sound-speed information
- echo measurements
- optical-system diagrams
- electromagnetic-spectrum reference data



## 1 WAVE IDENTIFICATION

Classify the following waves as: Transverse / Longitudinal / Electromagnetic / Mechanical (some may have more than one correct category).

| Wave           | Type(s) |
|----------------|---------|
| Water wave     | _____   |
| Sound in air   | _____   |
| Light wave     | _____   |
| Seismic P-wave | _____   |
| Radio wave     | _____   |
| Seismic S-wave | _____   |

## 2 WAVE CALCULATION

A communication signal has a frequency of 150 MHz and a propagation speed of  $3.0 \times 10^8$  m/s. Calculate its wavelength.

$$v = f\lambda$$

$f =$  \_\_\_\_\_

$v =$  \_\_\_\_\_

$\lambda =$  \_\_\_\_\_

## 3 UNDERWATER DETECTION

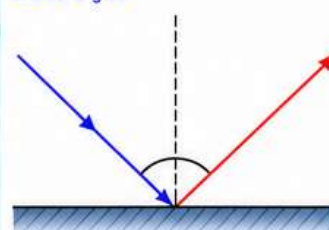
A sound pulse is sent toward an underwater object. The return time is 2.4 s. The speed of sound in water is 1 500 m/s. Calculate the distance to the object (account for the complete outward and return journey).



Distance = \_\_\_\_\_ m

## 4 REFLECTION SYSTEM

Complete the diagram and label: incident wave, normal, reflected wave and the angles.



## 5 OPTICAL DEVICE

Complete the ray diagram for a converging lens used in an imaging system. Describe the image produced.

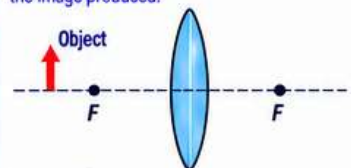


Image position: \_\_\_\_\_

Image type: \_\_\_\_\_

Image size: \_\_\_\_\_

Image orientation: \_\_\_\_\_

## 6 THERMAL DETECTION

Which region of the electromagnetic spectrum is best for detecting warm objects in darkness? Explain the physics behind your choice.



\_\_\_\_\_

## 7 COMMUNICATION NETWORK

Choose appropriate wave technology for each application. For each, give the wave or technology and one key property.

| Application                  | Wave/Technology | Property |
|------------------------------|-----------------|----------|
| Local wireless communication | _____           | _____    |
| Satellite communication      | _____           | _____    |
| High-speed optical data link | _____           | _____    |

## 8 RISK ASSESSMENT

Identify whether any proposed electromagnetic system requires additional exposure controls. Justify your answer scientifically.



**Safety matters.**

## 9 SYSTEM FAILURE

One communication signal cannot reach a receiver directly because of an obstacle (mountain). Propose a solution using reflection, diffraction or an alternative transmission method.



## FINAL PHYSICS DECISION

Which combination of wave technologies creates the most effective rescue network?

Your recommendation must use evidence from:

- one wave calculation
- one sound calculation
- wave behaviour
- optics
- electromagnetic-wave properties
- risk analysis

**Recommended system:**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_



## KEY TAKEAWAYS

- ✓ Different wave types have different properties and uses.
- ✓ Real-world problems often require more than one wave technology.
- ✓ Careful analysis and evidence lead to better decisions.
- ✓ Physics helps save lives.



## REAL-WORLD IMPACT

Wave technologies play a vital role in search and rescue, helping teams find people, communicate across difficult environments and gather critical information in challenging conditions.



## VOLUME 4 COMPLETE YOU CAN NOW:

- ☐ Describe transverse and longitudinal waves
- ☐ Measure amplitude and wavelength
- ☐ Calculate frequency and period
- ☐ Use the wave equation
- ☐ Explain reflection
- ☐ Explain refraction
- ☐ Explain diffraction
- ☐ Interpret wave graphs
- ☐ Explain sound transmission
- ☐ Relate frequency to pitch
- ☐ Analyse sound amplitude
- ☐ Calculate distance using echoes
- ☐ Construct light ray diagrams
- ☐ Analyse mirrors and lenses
- ☐ Explain colour and visible light
- ☐ Describe the electromagnetic spectrum
- ☐ Match electromagnetic waves to applications
- ☐ Evaluate electromagnetic-wave risks
- ☐ Analyse communication systems
- ☐ Solve integrated wave problems

